Embedded Fiscal Centralization: How Economic Openness Increased Fiscal Capacity and Redistribution in Contemporary China

Abstract

When can trade openness strengthen fiscal capacity? I argue that openness builds central fiscal capacity when export opportunities create new firms under a centrally controlled tax bureaucracy, limiting local influence over enforcement. China's World Trade Organization accession provides a most-likely test: the 2002 enterprise income tax (EIT) reform placed new firms under central collection and gave the center a share of income-tax revenue. Using a city-level Bartik design that instruments export exposure with tariff-predicted export growth, I find that exposed cities registered more private firms, had more frontline recruitment positions in the central tax bureaucracy, and generated more EIT revenue for the center. I also show that fiscal centralization enables selective compensation downstream: larger grants and greater fiscal dependence in cities where state-owned enterprise employment declined more. These findings suggest that the fiscal effects of openness depend on how governments organize tax collection and pool revenue across places.

Keywords: trade liberalization, fiscal capacity, redistribution, taxation, China

1 Introduction

When does trade openness build, rather than erode, fiscal capacity? Existing scholarship generally predicts that economic openness erodes public revenue. When tariff rates are high, tax bureaucrats can collect substantial revenue at visible border entry points without detailed information on firms' profits or household incomes. Trade liberalization can remove this dependable revenue before governments replace it with harder-to-administer value-added or income taxes (Baunsgaard & Keen, 2010; Cagé & Gadenne, 2018; Keen & Ligthart, 2002, 2005; Khattry & Rao, 2002). Yet export opportunities can also induce firms to enter the market, hire workers, and earn profits. This activity can expand the domestic tax base (Bachas et al., 2022). Openness may therefore destroy revenue at the border while creating tax potential behind it. Whether that potential strengthens the fiscal center depends on who controls the new tax source and how the resulting revenue is distributed.

I argue that trade openness builds fiscal capacity when export expansion brings new firms into a tax system that gives the center both a legal claim to tax revenue and control over collection. The registered entry of new firms raises the return to expanding the tax bureaucracy, while central control over collectors limits local bargaining over enforcement. Openness thereby expands the center's administrative reach and the revenue it can command. This expansion of central fiscal capacity also has distributive implications: pooling additional revenue across localities allows the center to pursue its own spending priorities, including compensation for politically salient groups.

Existing research offers important insights into the fiscal effects of openness but pays less attention to the political and administrative conditions that shape them. Work on trade and public finance explains tariff losses and domestic tax replacement, but offers a less developed account of how central–local conflict and control over tax bureaucrats shape revenue capture (Avi-Yonah, 2000; Bastiaens & Rudra, 2018; Boadway & Sato, 2009; Devereux et al., 2002; Gnangnon & Brun, 2019; Haque & Mukherjee, 2005; Mujumdar, 2004; Naito, 2006; Seelkopf et al., 2016; Slemrod, 2004; Sørensen, 2007). In contrast, research on fiscal capacity explains the institutional conditions under which governments can collect domestic revenue: information, enforcement, bu-

reaucratic organization, technology, and political authority can help transform tax claims into actual revenue (Besley & Persson, 2009; Bräutigam et al., 2008; Levi, 1988; Moore, 2015; Okunogbe & Tourek, 2024; Pomeranz, 2015; Slemrod, 2019; Tilly, 1992). In multilevel states, these institutional conditions can shape which government captures the revenue from openness.

Building on Ruggie's (1982) account of embedded liberalism, I call this process *embedded fiscal centralization*. Accounts of openness and compensation emphasize protection against market risk, primarily in industrialized democracies (Adserà & Boix, 2002; D. R. Cameron, 1978; Hiscox, 2001; Nooruddin & Rudra, 2014; Rodrik, 1998, 2000). In developing countries, however, greater redistribution first requires stronger tax collection. As trade taxes decline, governments must build the capacity to raise domestic revenue (Bastiaens & Rudra, 2018; Baunsgaard & Keen, 2010; Seelkopf et al., 2016). Fiscal centralization then allows the center to allocate these resources according to its political priorities.

I focus on China after World Trade Organization (WTO) accession as a most-likely case for the argument. Accession expanded opportunities for private firms and new entrants (Brandt et al., 2017; Fan, Li, & Yeaple, 2015; Gallagher, 2005; Heise et al., 2024). Established registration and reporting systems made new economic activity visible to tax authorities. China's highly centralized authoritarian system limited local officials' ability to resist the expansion of central fiscal authority (Y. Huang, 1996). Limited reliance on tariffs reduced the revenue at risk from liberalization.[1] The center also held statutory claims to revenue and exercised control over the central tax bureaucracy. Existing fiscal transfer mechanisms allowed the center to finance obligations administered by local governments, providing an established channel for responding to local fiscal pressures (Choi, 2016; Lou, 2008; Yang, 2004). These conditions made China especially favorable for examining how export expansion strengthens central fiscal capacity and enables redistribution.

Empirically, I follow Chinese prefecture-level cities from 2000 to 2007. A city-level Bartik measure combines pre-WTO industrial composition with national industry export growth. Because realized export growth may reflect domestic supply conditions, I instrument exposure with tariff-predicted export growth. The instrument uses product-level tariff reductions faced by Chinese ex-

porters and the same pre-WTO industry shares. Identification therefore comes from pre-treatment industrial composition interacted with common foreign tariff changes (Bartik, 1991; Bombardini & Li, 2020; Goldsmith-Pinkham et al., 2020; B. Li, 2018). To examine the downstream implications for redistribution, I turn to state-owned enterprise (SOE) restructuring, which gave rise to pressing demands for compensation during the decade. SOE restructuring began before WTO accession and had domestic causes (Hurst, 2009; Naughton, 2018). I therefore examine it separately, using a two-period city panel to assess fiscal compensation for places bearing concentrated adjustment costs.

I find that export exposure increased private-firm registrations and private-sector employment. Registrations also rose among foreign-invested firms, but the largest increase occurred among domestic private firms. In contrast, SOE entry did not change systematically. Central Tax Administration (CTA) recruitment also increased, mainly in county branches responsible for tax collection rather than city headquarters. The revenue results show both enterprise income tax (EIT) growth and a shift toward this tax. The center's EIT share, calculated from fiscal-book revenue using the legal sharing rules, is 48.4% higher across the exposure interquartile range. The increase in the center's value-added tax (VAT) share is smaller and statistically imprecise. EIT also accounts for a larger proportion of combined EIT and VAT revenue.

In addition, a ten-percentage-point decline in the SOE employment share predicts about 4.2% higher reported grant receipts. Cities with greater contraction also rely more on grant finance and record more retirees. Utility spending, other general spending, and public employment show no comparable response. The top quartiles of export exposure and SOE employment-share decline overlap only partially, consistent with national pooling across cities.

The article makes three contributions. First, it shows how central–local relations shape the fiscal gains from trade openness. Research on trade and taxation explains how tariff losses, domestic tax substitution, and tax competition affect government revenue (Bastiaens & Rudra, 2018; Baunsgaard & Keen, 2010; Cagé & Gadenne, 2018; Keen & Ligthart, 2005; Seelkopf et al., 2016). Whether an expanding domestic tax base strengthens the center, however, also depends on its abil-

ity to claim and collect revenue from that base. Central revenue claims can conflict with local governments' efforts to retain funds and protect firms, while local officials' promotion incentives shape fiscal extraction (Lü & Landry, 2014). These competing interests make control over tax collectors central to the fiscal consequences of openness. Building on research on fiscal rules and bureaucratic control (Beramendi & Rogers, 2020; Oates, 2005; Taliercio, 2004), I explain how statutory revenue claims and central supervision of collectors allow the center to capture revenue from trade-induced entry while limiting local influence over enforcement.

Second, I show how common national institutions generate uneven subnational state capacity. Comparative research emphasizes the territorial reach of bureaucratic rule (Garfias & Sellars, 2026; Levi, 1988; Mazzuca, 2021; O'Donnell, 1993; Véjares, 2025). I identify an economic source of variation in subnational fiscal capacity: changes in the location of taxable activity alter the center's incentives to assign bureaucratic personnel. Export expansion increased potential returns to tax administration more in some cities than in others. The frontline tax bureaucracy expanded recruitment where the registered tax base grew. Uneven bureaucratic expansion can therefore emerge within a common national hierarchy as the center responds to differences in local revenue opportunities. This finding connects trade-induced economic change to the territorial development of fiscal capacity, showing how trade can shape the subnational distribution of bureaucratic resources.

Third, I show how fiscal centralization enables selective compensation under openness and can deepen recipient localities' dependence on the center. Accounts of embedded liberalism explain responses to trade-related risks (Adserà & Boix, 2002; Rodrik, 1998; Ruggie, 1982) but often assume the capacity to finance them. Institutions that capture and pool revenue make cross-regional redistribution possible. Such redistribution can also deepen recipient localities' dependence on the center (Rodden, 2002). In China, greater SOE contraction was associated with more grants from the central government. This pattern accords with arguments that authoritarian rulers favor spatially concentrated, administratively visible constituencies with the organizational capac-

ity to press their demands (Albertus, 2024; R. M. Desai et al., 2009; X. Huang, 2020; Nooruddin & Rudra, 2014; Pan, 2020).

In what follows, Section 2 develops the argument and its two principal observable implications. Section 3 explains the choice of China and the relevant institutions. Section 4 presents the empirical strategy. Sections 5 and 6 report the revenue and fiscal compensation evidence. Section 7 concludes.

2 Theory

In this section, I first explain why registered-firm entry increases demand for tax personnel, then examine central control over collection and the political priorities shaping redistribution in authoritarian contexts.

2.1 Trade Openness and the Taxable Domestic Base

Trade liberalization affects public revenue through two channels. Reductions in domestic tariffs erode tariff revenue, which lower-capacity tax systems often struggle to replace (Baunsgaard & Keen, 2010; Cagé & Gadenne, 2018; Keen & Ligthart, 2002, 2005; Khattry & Rao, 2002). Improved access to foreign markets, by contrast, can create taxable domestic activity. Exporters expand production and hire workers, while suppliers and related businesses grow around them. These gains may arise even when they do not fully offset lost border revenue (Baunsgaard & Keen, 2010; Cagé & Gadenne, 2018). I focus on this second channel: whether foreign market access generates activity that governments can identify and tax.

The domestic tax base expands at two margins in response to an export boom. At the intensive margin, existing taxpayers can increase output, payrolls, profits, and transactions (Chaney, 2008; Melitz, 2003), potentially raising their tax liabilities. At the extensive margin, registered-firm entry adds new taxpayers. In developing economies, trade liberalization often removes domestic restrictions on which firms can trade. Where trading rights were reserved for

selected firms, opening access allows more businesses to enter export production and related activities. Openness can therefore encourage firm entry by removing domestic restrictions as well as foreign trade barriers.

Expansion through entry is especially likely to require additional administrative resources because each entrant adds a separate unit to administer. Registration gives the firm a legal identity and recorded location. Accounts, invoices, payrolls, and reporting obligations make its activity visible to revenue authorities (Bachas et al., 2022). Collectors must classify each new taxpayer, assess its liabilities, monitor compliance, and collect payments. By comparison, higher output from an existing firm can be handled more easily within an established administrative relationship. Registering a new firm adds another administrative relationship, increasing administrative demands as well as potential revenue. Where the center has a legal claim to tax revenue, the entry of registered firms increases its expected return from allocating additional administrative resources to those localities.

If export-driven entry occurs mainly in the informal sector, tax collection becomes more difficult because many new firms remain outside official records (Ulyssea, 2018). This problem is less likely where even a moderately developed registration system brings new firms into those records, although broad exemptions can still leave their income outside the effective tax base (Abramovsky et al., 2014). Registration, accounting, and reporting systems allow collectors to identify firms and their obligations even when enforcement is incomplete (Kleven et al., 2016; Pomeranz, 2015). However, identifying registered firms does not by itself generate revenue. Turning this information into collections requires administrative capacity to verify liabilities and enforce payment (Okunogbe & Tourek, 2024).

The center's ability to capture this revenue depends on who controls collection. An expanding tax base raises the stakes of central–local bargaining over enforcement and revenue.

2.2 Central–Local Conflict over Tax Collection

Central and local governments both need revenue to meet spending obligations. The center finances national priorities, such as infrastructure construction and redistribution across regions. Local governments provide public goods and fund development projects within their jurisdictions. These competing needs make conflict over tax collection and retained revenue common in multilevel states (Rodden, 2002, 2004). Consolidating fragmented tax jurisdictions helped build national revenue in Europe (Dincecco, 2009). In democratic Argentina and Brazil, national executives reasserted fiscal control by changing revenue sharing and subnational autonomy (Eaton & Dickovick, 2004). Fiscal centralization gives the center greater authority over tax bases and the division of shared tax revenue, even when spending responsibilities remain decentralized. Its extent depends on who sets tax rules, supervises collectors, and receives the revenue.

Registered-firm entry brings new revenue into this central–local conflict. The center seeks stable revenue for national priorities and less dependence on subnational remittances. Local governments seek to retain revenue while attracting investment and protecting employment. Their proximity to taxpayers can facilitate registration and information sharing, but it can also enable protection of favored firms, tolerance of underreporting, and influence over tax treatment. Without a centrally controlled tax bureaucracy, a formal central claim over tax revenue can consequently coexist with selective local implementation of national rules (Garfias & Sellars, 2022, 2026; Holland, 2016; Mazzuca, 2021; Véjares, 2025).

Firms can seek favorable treatment through political connections (H. Cai & Liu, 2009; H. Li et al., 2008). Local officials may protect them and tolerate some tax avoidance in return for investment or employment. On the other hand, officials may also use their discretion to extract excess resources (Liu, 2018). In either case, tax treatment and enforcement become objects of bargaining (Holland, 2016). Monitoring firm registration alone cannot secure this revenue for the center. Registration identifies potential taxpayers, but collecting from them requires staff who verify liabilities, investigate underreporting, and enforce payment (Basri et al., 2021; Okunogbe & Tourek, 2024; Slemrod, 2019). Where local officials influence these staff members, better taxpayer

records can coexist with selective enforcement (Choi, 2016). The center therefore needs a central tax bureaucracy with both the capacity to collect and the authority to enforce national tax rules despite local pressure.

Notably, a tax office's location does not determine its political accountability. The key is bureaucratic control. Collectors working in a local economy can answer to the center when it controls their careers, reporting, taxpayer records, and enforcement priorities. These arrangements reduce collectors' dependence on local coalitions and limit local officials' ability to withhold information or weaken collection (Levi, 1988; Okunogbe & Tourek, 2024; Pomeranz, 2015; Slemrod, 2019; Taliercio, 2004). Central supervision makes enforcement of national rules less dependent on local political interests (Garfias & Sellars, 2026; Taliercio, 2004).

Comparative research highlights the importance of central bureaucratic control for administrative effectiveness and the implementation of central priorities (Cantoni et al., 2024; Jacoby et al., 2021; Malesky et al., 2014). For tax collection, this implies that central control over personnel and enforcement helps protect revenue collection from local interference (Jia & Nie, 2017; Kong & Liu, 2024; Zhang et al., 2024).

Importantly, political centralization is a key scope condition for central revenue capture. The center needs sufficient authority over local governments to protect tax collection from local interference. Control over local officials' appointments, promotion, and discipline can make local resistance to national rules more costly (Blanchard & Shleifer, 2001; Y. Huang, 1996). Where the center has little leverage over subnational officials, local governments can obstruct collection or shield favored firms despite formal central tax authority. An expanding registered tax base is then less likely to yield revenue under central control.

For the center to capture revenue from the export boom, central control must therefore extend to the local offices that assess and collect taxes from an expanding taxpayer population. Deploying centrally accountable collectors in localities where new firms register gives the center the personnel to act on taxpayer information and limits local bargaining over enforcement. It

should be noted that the specific agencies and taxes involved may depend on the institutional setting.

2.3 National Revenue Pooling and Selective Redistribution

Central revenue capture provides the fiscal basis for national revenue pooling: the center can use revenue raised in one locality to finance obligations elsewhere. Pooling matters because the places generating additional tax revenue need not be those facing the greatest demands for support. Economic contraction or domestic restructuring may increase social obligations in localities whose own tax bases are weak or declining. By drawing on revenue generated elsewhere, the center can finance support beyond what these localities could sustain from their own resources (Adserà & Boix, 2002; Beramendi & Rogers, 2020; Oates, 2005; O'Brien et al., 2025; Rodden, 2002; Rodrik, 1998).

The center must still decide which groups and places receive support. Concentrated trade losses generate demands for protection, redistribution, or social insurance (Autor et al., 2013, 2020; Broz et al., 2021; Colantone & Stanig, 2018). In democracies, electoral competition can direct support toward broad constituencies through insurance, subsidies, or legislated transfers (Adserà & Boix, 2002; Hiscox, 2001; Rodrik, 1998). Revenue constraints, administrative capacity, and intergovernmental bargaining shape that response. Recent research on trade conflicts and the backlash against globalization underscores the political consequences of decisions about whom to protect (Broz et al., 2021; Farrell & Newman, 2023; Goldberg, 2023; L. Jiang et al., 2023; Zeng & Kim, 2025).

Authoritarian rulers face weaker electoral pressure for broad redistribution and have greater discretion to target groups and places (X. Huang, 2020; Pan, 2020). For them, a constituency's capacity for collective action matters more than hardship alone (Chen et al., 2016; R. M. Desai et al., 2009). Spatial concentration, workplace and neighborhood networks, and shared organizational ties lower coordination costs for the affected groups. They make protest and disruption more credible (H. Zhu, 2025).

Organized groups can make economic losses politically costly for rulers. They can coordinate demands for support and hold the regime responsible for failing to address their hardship. Targeted redistribution can help rulers defuse these organized challenges (Albertus, 2024), particularly when repression is sufficiently costly. Rulers should therefore be more responsive to groups with greater capacity for collective action. When these groups are geographically concentrated and their benefits are administered locally, their demands create pressure for grants to those jurisdictions. In contrast, equally severe losses among diffuse or weakly organized groups may elicit less support because their capacity for collective disruption is weaker.

Where social obligations are administered locally, grants from higher levels of government provide a channel through which pooled revenue can finance support for politically consequential groups. They enter local budgets and can finance social insurance, unemployment relief, income support, adjustment programs, and other commitments (Nooruddin & Rudra, 2014; Wong & Ngok, 2006). Governments may also provide public employment or other benefits to relatively privileged losers (Nooruddin & Rudra, 2014). Transfers can replace revenue lost during contraction and allow local governments to maintain services. As a result, where organized constituencies exert political pressure for support, grants should rise as local resources weaken or obligations grow. Localities become more fiscally dependent when grants rise relative to locally controlled revenue or spending.

2.4 Observable Implications

The argument yields two principal implications. The first concerns central revenue capture from expanding market opportunities. The second concerns a downstream consequence: national pooling makes centrally captured revenue available to finance locally administered obligations according to the center's political priorities.

Central revenue capture. Localities more exposed to improved foreign market access should generate more central revenue where the center holds statutory claims over domestic taxes and con-

trols tax collectors (Falleti, 2010; Taliercio, 2004). In response to export-driven growth, registered-firm entry and recruitment of tax collectors accountable to the central government should both increase.

Fiscal compensation. National revenue pooling should increase redistribution toward localities where politically organized groups press for support. These localities should receive larger grants and become more fiscally dependent, with transfers accounting for a greater share of their spending. Revenue captured from export expansion can therefore finance obligations arising from economic distress. This prediction is more likely to apply where local governments administer these obligations and the center has discretion over grants (Oates, 2005; Rodden, 2002).

3 Post-WTO China as a Most-Likely Case

Post-WTO China offers a most-likely case for examining central revenue capture and fiscal compensation in response to an export boom. The center combined statutory claims over domestic taxes with a centrally supervised tax bureaucracy operating in local economies, giving it the means to capture revenue from an expanding registered-firm tax base. Grants from higher levels of government, in turn, historically helped local governments finance their social obligations. The section reviews central tax administration and the 2002 EIT reform, private-firm entry after WTO accession, and the fiscal burdens of SOE restructuring, then sets out the specific implications for China.

3.1 Central Tax Infrastructure and the 2002 EIT Reform

China entered the WTO in December 2001 with a tax system divided between central and local authorities. The 1994 tax-sharing reform had expanded central fiscal authority after decentralization in the 1980s and early 1990s, bringing some revenue streams under central control and strengthening collection and monitoring (Choi, 2016). It established two parallel bureaucracies: the centrally controlled CTA and the locally controlled Local Tax Administration (LTA) (Y. Huang, 1996;

Lou, 2008; Yang, 2004). The CTA gave the center a network of local offices, but its workforce remained relatively small and local governments retained substantial fiscal authority. Figure 1 summarizes this dual structure.

Under this arrangement, local governments controlled EIT collection from private and locally affiliated firms before 2002, while the CTA administered central and shared taxes, including VAT. Local officials could relax EIT enforcement to protect investment. Capturing revenue from private-sector growth therefore required extending the CTA's jurisdiction to the new firms created by the export boom.

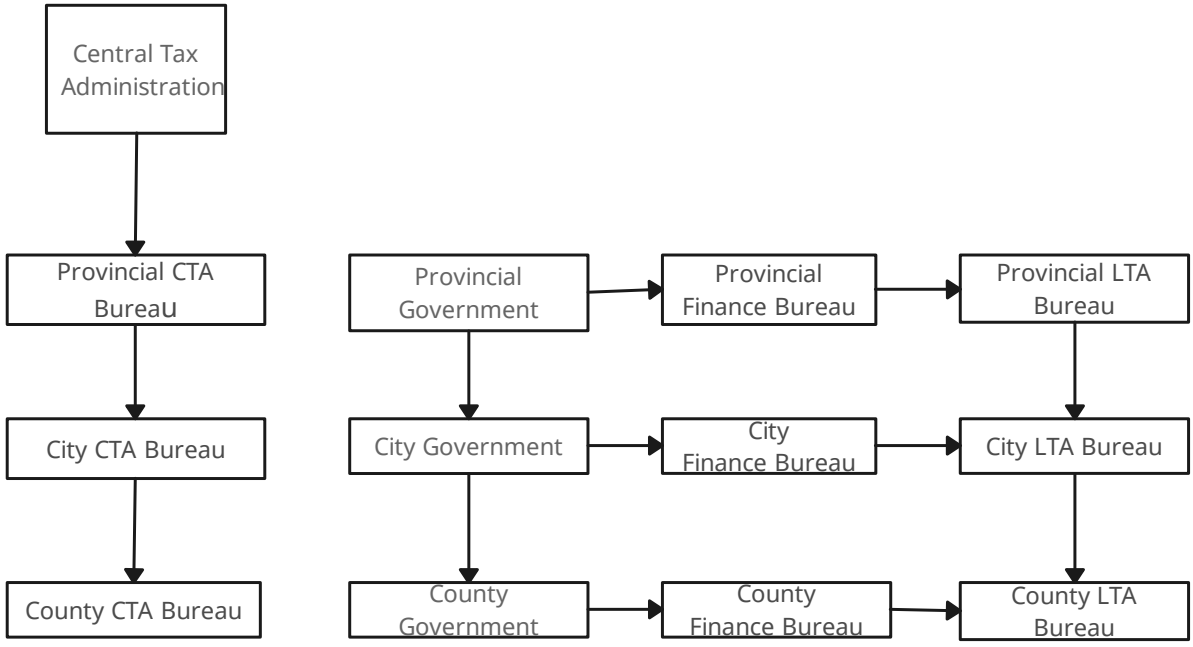


Figure 1. China's tax administration, 1994–2017

Note. Arrows denote formal authority over revenue targets, hiring, promotion, other personnel decisions, and funding. The figure ends in 2017 because the 2018 reform merged the CTA and LTA systems at and below the provincial level, ending the dual structure shown here.

The 2002 EIT reform expanded both the center's revenue share and the CTA's collection jurisdiction. It initially divided income-tax revenue equally between the center and local governments; from 2003 onward, the center received 60%. It also placed most newly established firms under CTA collection, while many pre-2002 firms retained their previous collecting agency (State

Administration of Taxation, 2002; State Council, 2001).[2] The reform protected each locality's 2001 income-tax baseline and directed additional central revenue to general transfers, mainly for central and western regions (State Council, 2001). Baseline protection limited local fiscal losses and preserved incentives to attract investment and exporters.

Central supervision of CTA personnel made collection from new firms less dependent on local officials. Each eligible entrant expanded the CTA-administered tax base, while the center's 60% share raised its return to staffing and enforcement. With these rules fixed nationally, variation in local export opportunities provides a way to examine where the center expanded its collection capacity.

VAT provides a useful comparison because, unlike EIT, its collection jurisdiction and sharing rules were unchanged by the 2002 reform. It is a major domestic revenue source and an important substitute for declining tariff revenue (Buettner & Madzharova, 2018; Keen & Lockwood, 2010). In China, the CTA had administered VAT since the 1994 reform (Choi, 2016), and computerized invoice verification was being expanded through the Golden Tax Project (State Administration of Taxation, 2000). Export growth could increase revenue from both taxes, so I examine how revenue from each tax responded and whether export exposure shifted the composition of revenue toward EIT.

3.2 WTO Accession and Private-Firm Entry

WTO accession expanded opportunities for private firms. Before accession, trading rights were primarily restricted to large SOEs and foreign-invested enterprises (FIEs). Most private firms could not trade directly and relied on state-owned or foreign intermediaries. Under commitments negotiated with its trading partners, China dismantled this permit barrier within three years of WTO accession. By 2004, private firms could reach foreign markets directly (Brandt et al., 2017; Gallagher, 2005; Naughton, 2018). Accession also reduced trade barriers and uncertainty about market access (Fan, Li, & Yeaple, 2015; Heise et al., 2024). Private enterprises (PEs), previously the most restricted ownership group, benefited particularly from these changes.

Growth extended beyond incumbent exporters. New PEs entered export production, supplier networks, logistics, processing, and related services. SOEs and many FIEs already had established state relationships and export channels. Private firms gained disproportionately from expanded trading rights and more certain access. Their weaker ties to central administrators also made tax evasion more likely. Registration and assignment to a collecting agency nevertheless brought most of these new firms under central tax administration.

Appendix Figures A1 and A2 summarize exports, new registrations, and EIT revenue across these years. The domestic export boom created identifiable firms, records, and physical establishments. The CTA hierarchy and the 2002 collection and sharing rules made this expanding base available to central tax administration (Boone, 1994; Okunogbe & Tourek, 2024; Pomeranz, 2015).

3.3 SOE Restructuring

In China in the early 2000s, SOE restructuring created local fiscal burdens that arose from domestic reforms distinct from WTO accession. From the late 1990s into the mid-2000s, the government closed, merged, and privatized inefficient enterprises, hardened budget constraints, and routed tens of millions of workers through *xiagang* and related layoff programs (Hurst, 2009; Naughton, 2018). Trade liberalization expanded opportunities for private employment in China, while SOE restructuring generated the most politically salient demands for compensation. The compensation examined here therefore concerns the social costs of a distinct process of domestic reform, rather than job losses caused by trade liberalization. China's case differs from settings where import competition generates the main demands for compensation.

Laid-off SOE workers formed a politically salient constituency with inherited claims on the state. Public employment had bundled wages with housing, pensions, medical care, and status (H. Y. Lee, 2000; Luo et al., 2023). Losses were concentrated in older industrial cities, where factory neighborhoods and work-unit networks facilitated petitions and protests (Y. Cai, 2002; Frazier, 2004; Friedman, 2014; Hurst & O'Brien, 2002; C. K. Lee, 2007). Rural residents and mi-

grants, by comparison, lacked the same claims rooted in prior state employment (C. K. Lee, 2007, pp. 12, 22–23). I focus on SOE workers because their losses were concentrated, their claims invoked public commitments, and their employment decline and fiscal burdens are measurable at the city level.

Appendix Figure A4 provides evidence on these demands. In the 2003 Chinese General Social Survey (CGSS), respondents laid off from public-sector jobs expressed stronger redistributive preferences and gave higher priority to reducing unemployment. They also more often reported declining status and worsening living conditions. Figure 2 shows the sharp fall in SOE manufacturing employment and the provincial association between rising unemployment and subsequent net transfers.

After 1994, local governments remained responsible for major spending programs and social stability, while the center rebuilt revenue authority through the CTA (Y. Huang, 1996; Lou, 2008; Tsui, 2005; Yang, 2004). Restructuring widened the gap between local resources and obligations. As their tax bases contracted, older industrial cities faced pension arrears and enterprise closures, along with obligations to provide unemployment relief, minimum-livelihood guarantees, reemployment assistance, and support for retirees (Hurst, 2009; Naughton, 2018; Solinger, 2002; State Council, 2000). Higher-level transfers financed these commitments and offset lost revenue.

The 2002 reform directed additional central income-tax revenue toward general transfers, principally to central and western regions (State Council, 2001). Alongside the social-policy rules for pensions and unemployment assistance, this provision gave the center a means to finance locally administered commitments (Lou & Wang, 2008). Local governments' obligations took several forms, including unemployment relief and minimum-livelihood guarantees (State Council, 2000), pension payments and support for retirees (Hurst & O'Brien, 2002), and reemployment assistance (P. Li & Zhang, 2004).

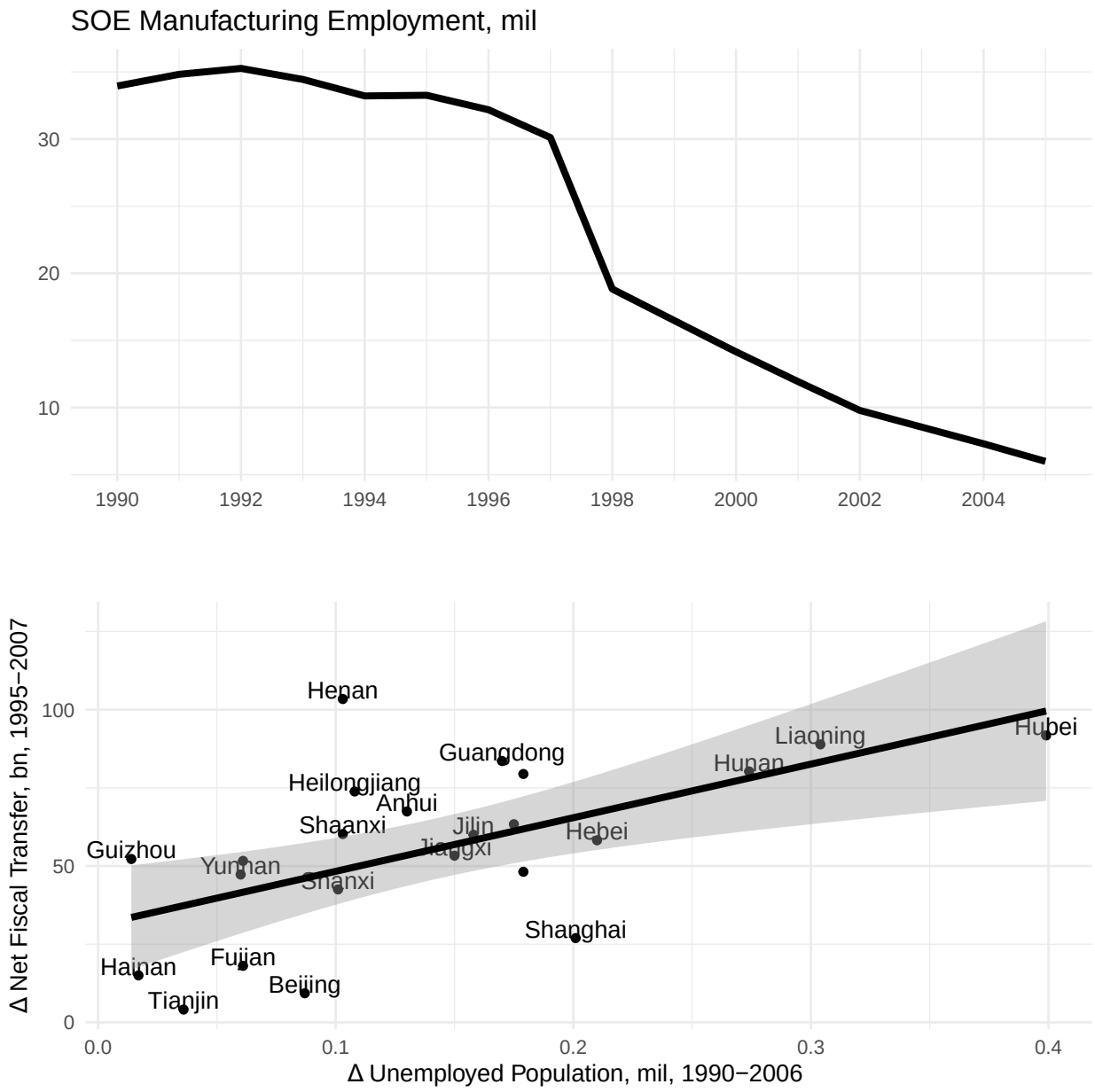


Figure 2. SOE contraction, unemployment, and fiscal transfers, 1990–2007

Note. The upper panel reports SOE manufacturing employment during 1990–2005. The lower panel plots the 1995–2007 change in provincial net fiscal transfers against the 1990–2006 change in unemployment. The transfer change is expressed in billions of Chinese yuan (CNY). Data source: *China Statistical Yearbook*, various years.

3.4 Implications for China

I first examine firm entry through private-firm registrations. Private firms should respond most strongly to export expansion because they benefited particularly from expanded trading rights. I then examine CTA recruitment positions, which measure the accompanying bureaucratic adjustment. Recruitment should be concentrated in county branches responsible for tax collection. Finally, I examine revenue capture through fiscal-book EIT and the center's EIT share. The 2002 reform changed EIT collection and sharing while leaving VAT rules unchanged. Export expansion should increase fiscal-book EIT and the center's EIT share. Whether EIT revenue grows relative to VAT revenue provides additional evidence on the collection mechanism.

I then examine fiscal compensation by testing whether cities with larger declines in the SOE employment share received more grants from higher levels of government. Their budgets should also reflect greater spending obligations to SOE workers. Dependence on grant finance should rise when grants increase relative to spending. I also show where export-related revenue growth and fiscal compensation occurred across cities.

4 Data and Empirical Strategy

The design follows prefecture-level cities from 2000 to 2007. The baseline predates WTO accession, the export boom, and the 2002 EIT reform. The endpoint allows entry, recruitment, and revenue effects to emerge before the Global Financial Crisis and 2008 stimulus (Bombardini & Li, 2020; B. Tian, 2014).

Prefecture-level cities are the lowest administrative units with consistent economic, employment, and transfer data (Bombardini & Li, 2020). They sit between provinces and counties and typically administer an urban core and surrounding counties. Their governments coordinate development and budgetary obligations and interact with central and local tax offices. City jurisdictions encompass industrial districts, firm registration authorities, CTA branches, and governments administering pensions, unemployment relief, and public services. I aggregate fiscal accounts for

each city and its counties so that revenue, spending, and economic activity are measured for the same prefecture-level city.

4.1 Export Exposure

I measure each city's post-WTO export exposure by combining its pre-accession industrial composition with national industry export growth from 2000 to 2007. This shift-share design builds on Bartik and related approaches to estimating the local effects of China's WTO accession and trade shocks (Autor et al., 2016; Baccini & Weymouth, 2021; Bartik, 1991; Bombardini & Li, 2020; Campello & Urdinez, 2021; B. Li, 2018; Rettl, 2025; Scheve & Serlin, 2023). Figure 3 shows the industry-level export expansion underlying the measure.

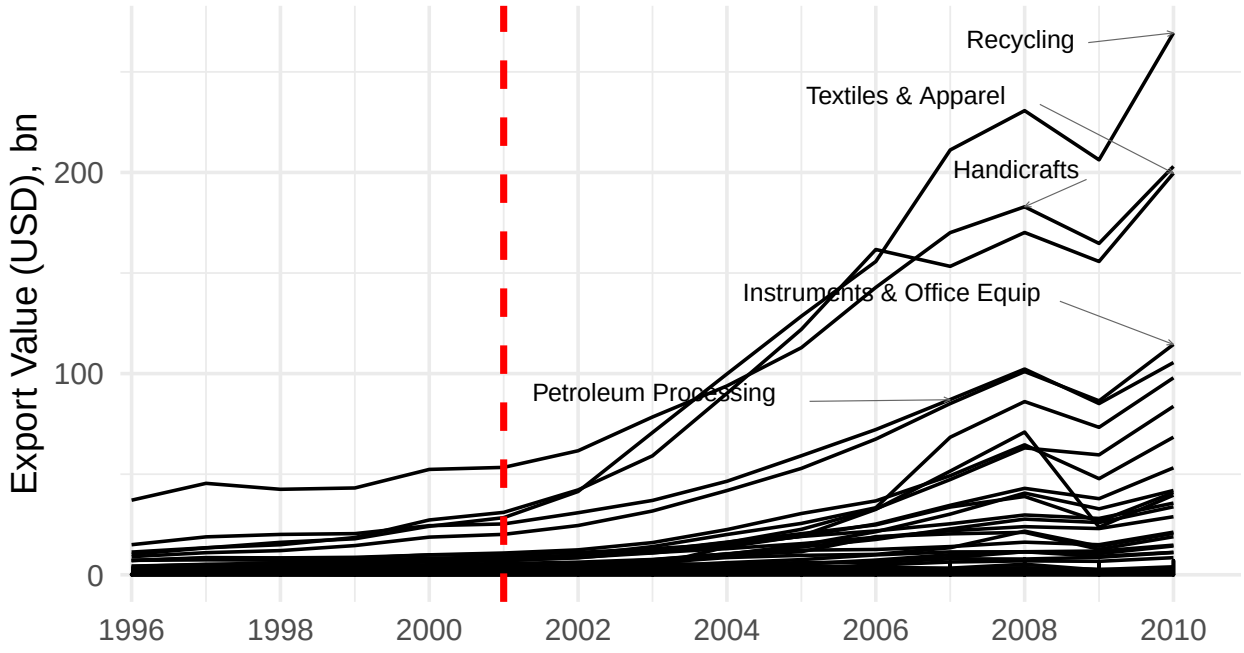


Figure 3. Industry-level export values, 1996–2010

Note. Annual export values are shown for all two-digit industry groups. The dashed line marks China's WTO accession in December 2001. Leader lines connect the five labeled industries to their respective series.

The exposure measure is

$$EXPShock_i^{07} = \sum_k \frac{Y_{ik}^{00}}{Y_k^{00}} \cdot \frac{\Delta EXP_k^{00-07}}{Y_i^{00}}, \tag{1}$$

where Y_{ik}^{00} is city i's industrial output in industry k in 2000, Y_k^{00} is national output in that industry, Y_i^{00} is city output, and ΔEXP_k^{00-07} is national export growth in industry k. Export exposure is measured in US dollars per thousand yuan of baseline industrial output. I use industrial output shares because the mechanism operates through changes in the taxable production base rather than labor-market exposure. As in other shift-share designs, the identifying assumption is that, absent post-WTO industry export shifts, cities with different baseline industry shares would not have experienced differential trends in firm entry, CTA recruitment, or fiscal outcomes (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020).

Cities concentrated in industries with large subsequent export gains receive greater exposure. The weights fix industrial composition before accession, so the measure does not use contemporaneous local performance to assign exposure. The resulting comparison resembles difference-in-differences across cities with different initial industry portfolios (Scheve & Serlin, 2023).

Realized national export growth can still reflect domestic supply conditions, industry policy, and other changes affecting entry, tax administration, and fiscal outcomes (Borusyak et al., 2022; Goldsmith-Pinkham et al., 2020). I therefore instrument Bartik exposure with tariff-predicted export growth, following Bombardini and Li (2020) and B. Li (2018). The instrument predicts industry growth from product-level tariff reductions faced by Chinese exporters. It then aggregates that predicted component using the same pre-WTO output shares:

$$\widetilde{EXPShock_i}^{07} = \sum_k \frac{Y_{ik}^{00}}{Y_k^{00}} \cdot \frac{\widetilde{\Delta EXP_k}^{00-07}}{Y_i^{00}}, \tag{2}$$

where $\widetilde{\Delta EXP_k}^{00-07}$ is tariff-predicted national export growth in industry k. Appendix F details its construction. Foreign tariff cuts changed market access for Chinese industries and were not chosen by Chinese cities. Combining these changes with pre-WTO industrial composition is intended to

isolate externally driven exposure from contemporaneous local growth, fiscal policy, or administrative capacity (Kovak, 2013; Y. Tian, 2024). Appendix H.2 examines Rotemberg weights, cumulative industry exclusions, alternative exposure measures, geographic restrictions, and province fixed effects. Appendix H.5 reports pre-treatment tests.

4.2 Data and Outcomes

Export data come from United Nations (UN) Comtrade and are concorded to Chinese industries; output shares come from the *Annual Survey of Manufacturing Firms* (ASMF).[3] Tariff rates come from MAcMap-HS6 (Guimbard et al., 2012). Fiscal outcomes come from the *National Prefecture and County Finance Statistics Compendium*, and SOE employment shares come from the ASMF. Baseline controls come from the 2000 census and city yearbooks. Appendix Table A5 summarizes the variables. Appendix H.4 reports transformed and per-capita specifications.

The outcome data measure firm entry, CTA recruitment, fiscal revenue, and grant receipts. Firm registrations measure growth in the population of taxable firms; CTA recruitment positions measure demand for personnel to administer them. I measure firm entry using State Administration for Industry and Commerce registrations, aggregated by city and ownership type between 2000 and 2007.

I collect original data on CTA recruitment from national civil-service examination records. For each recruitment listing, I record the location, hiring office, job description, qualifications, and number of positions. These counts measure CTA recruitment positions rather than the stock of employees.[4] I distinguish branch offices, which register taxpayers, collect taxes, and enforce tax obligations, from headquarters offices responsible for general administration.

I use the same export-exposure measure to test for growth in taxable firms, frontline tax recruitment, and central revenue. Fiscal accounts provide the tax revenue used to calculate the center's EIT and VAT shares.

The main fiscal outcomes are the center's EIT share, its VAT share, and EIT's share of combined EIT and VAT revenue. The first two measures are revenue amounts accruing to the

center. The third is the percentage of combined EIT and VAT revenue that comes from EIT. I also examine the log ratio of the center's EIT share to VAT and consumption-tax payments to the central government. I calculate the central shares using the legal sharing rules. In 2007, the center received a 60% share of income-tax revenue. Fiscal books report the local 40%, so the center's EIT share equals 1.5 times fiscal-book EIT. VAT was shared 75:25, making the center's VAT share three times the fiscal-book amount. The CTA already administered VAT and consumption tax before 2002.

VAT's collection and sharing rules did not change in 2002. Comparing EIT with VAT therefore helps assess whether export exposure increased revenue disproportionately from the tax affected by the reform. Most new firms fell under CTA jurisdiction in this period. All tax outcomes exclude rebates and other receipts from higher levels of government. Appendix H.3 documents construction, composition, and outlier checks.

Grants received from higher levels of government comprise tax rebates, general and special transfers, and other subsidies. I harmonize these receipts across the 2000 and 2007 fiscal schedules. Total grants measure resources entering local budgets, and per-capita grants measure their intensity relative to population. Fiscal dependence is the log ratio of grants to spending. Social-security spending and recorded retirees capture obligations from SOE restructuring. Together, these measures test whether fiscal compensation and public obligations rose in cities where the SOE employment share declined. Appendix H.6 documents the grant measure and reports no-control and other-fiscal-outcome checks.

4.3 Empirical Specifications

The export-exposure analysis estimates the following cross-sectional two-stage least-squares (2SLS) equation:

$$Y_i = \beta_1 EXPShock_i^{2007} + \beta_2 X_i + \theta_r + \varepsilon_i, \tag{3}$$

where Y_i denotes a firm-entry, CTA-recruitment, or fiscal outcome. I instrument $EXPShock_i^{2007}$ with tariff-predicted export exposure. The vector X_i contains controls measured in 2000, and θ_r

denotes regional fixed effects. Preferred controls are the baseline share of the population employed in manufacturing, the unemployment share, the share of the population aged 20–34, the male share, and the first three principal components of industrial structure. Private-sector growth is a post-treatment mechanism outcome and therefore does not enter the controls. Downstream GDP, population, and firm growth likewise remain outside the preferred models.

Tariff-predicted exposure strongly predicts realized exposure. For each outcome, I estimate the first stage using the corresponding sample and the full set of controls, with standard errors clustered by province. The effective F-statistics equal 163.6 for private-firm entry and range from 133.8 to 167.6 across the main firm-entry, CTA-recruitment, and fiscal outcomes.

The instrumental-variable (IV) design estimates the effects of tariff-induced export exposure on private-firm registration, frontline recruitment, fiscal-book EIT, and the center's EIT share. I compare effects across firm ownership types, CTA office levels, and taxes with different collection and sharing rules.

Private firms benefited disproportionately from WTO-era access, and most new firms came under CTA jurisdiction in or after 2002. County-branch recruitment measures positions for taxpayer registration, tax collection, and enforcement. Headquarters recruitment measures positions in general administration. EIT's share of combined EIT and VAT revenue captures changes in the relative contribution of the two taxes. The ratio of the center's EIT share to VAT and consumption-tax payments to the center provides a second comparison. VAT's established central administration makes it a useful benchmark. The IV estimates identify effects of export exposure on each outcome. Stronger responses in collection branches and in EIT would support the argument that openness strengthened the center's ability to tax new firms.

The robustness analyses address distinct concerns about this comparison. Because initial industrial composition may also reflect earlier development paths, I examine whether predicted exposure is associated with fiscal changes before WTO accession. To assess dependence on particular industries, I calculate Rotemberg weights and reconstruct both realized exposure and its instrument after sequentially removing the most influential industries. Excluding coastal areas and

major administrative centers tests whether the findings depend on cities with distinctive access to trade or political resources. Province fixed effects provide a comparison among cities within the same province.

For fiscal compensation, I estimate a two-period city panel from 2000 to 2007:

$$Y_{it} = \beta_1 SOE_SHR_{it} + \beta_2 X_{it} + \theta_i + \gamma_t + \varepsilon_{it}, \tag{4}$$

where Y_{it} denotes a grant, spending, retiree, or fiscal-dependence outcome. The variable SOE_SHR_{it} is SOEs' share of employment in the ASMF firm sample, and X_{it} contains time-varying controls. θ_i and γ_t denote city and year fixed effects, respectively. Because the regressor measures the remaining share, negative coefficients indicate larger outcomes where SOE employment shares declined further. Grants measure the scale and intensity of fiscal compensation, fiscal dependence measures reliance on grant finance, and spending and retirees capture restructuring-related obligations.

In the cross-sectional export analysis, each city appears once, making city clustering equivalent to heteroskedasticity-robust inference. Main tables cluster standard errors by province to allow dependence within shared policy and economic environments (A. C. Cameron & Miller, 2015). Appendix H.1 reports unclustered results and provincial wild-cluster bootstrap checks (A. C. Cameron et al., 2008; MacKinnon et al., 2023). Appendix H also examines transformations, per-capita scaling, joint ownership-sector models, timing, and placebos. The two-period SOE models cluster standard errors by city.

5 Export Exposure and Central Fiscal Capacity

5.1 Private-Firm Entry

Table 1 reports the estimated effect of export exposure on firm registration. The argument predicts the strongest response among PEs, which had been more constrained by the pre-WTO trading-rights regime. FIEs and SOEs were already more integrated into global trade networks.

Model 1 reports the first stage. Models 2–4 estimate effects on new private-firm registrations while progressively adding baseline controls constructed using principal component analysis (PCA) and census covariates. Models 5 and 6 report inverse-hyperbolic-sine (IHS) and per-capita specifications for private-firm registrations, respectively. The remaining columns report private-sector employment and level changes in FIE and SOE registrations.[5]

Export exposure produces large, robust increases in private-firm entry across count, IHS, and per-capita specifications, alongside higher private-sector employment. In the full-control count model, a one-standard-deviation increase of 55.6 exposure units implies roughly 460 additional private firms between 2000 and 2007 (Appendix Table A5). The corresponding increase is roughly 15 FIEs on the same count-change scale. SOE entry does not respond systematically. Figure 4 shows the residualized descriptive correlation.

The comparison across ownership types shows the largest increase in registrations among domestic private firms, which faced the greatest constraints before accession and whose new entrants generally came under CTA collection in or after 2002. The estimates thus show that export-driven entry expanded the population of firms covered by the new collection rules.

Table 1. Export Exposure and Firm Registration by Ownership Type

	First Stage	Δ Private Firms					Δ PE Emp	Δ FIE	Δ SOE
	(1)	(2)	(3)	(4)	IHS	Per Cap.	Emp.	Count	Count
Export Exposure		10.407***	11.701***	8.247***	0.007***	0.008***	1.144**	0.278**	0.038
		(2.812)	(2.954)	(2.331)	(0.002)	(0.002)	(0.420)	(0.129)	(0.104)
$\widetilde{\text{Export Exposure}}$	1.679***								
	(0.105)								
Region FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
PCA Ind		N	Y	Y	Y	Y	Y	Y	Y
Controls		N	N	Y	Y	Y	Y	Y	Y
First-stage F	255.0	239.2	205.9	163.6	163.6	163.6	133.8	163.6	163.6
Num. obs.	312	310	310	310	310	310	251	310	310

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: export exposures are measured in US dollars per thousand yuan of baseline industrial output; firm outcomes are counts, except that private-sector employment is measured in thousands and the per-capita column is expressed per 10,000 baseline residents. Transformations: the column labeled IHS uses inverse hyperbolic sine; private-sector employment is divided by 1,000; the per-capita column reports private-firm changes divided by baseline population and multiplied by 10,000; FIE and SOE columns report level count changes, matching the private-firm count outcomes. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure; columns marked PCA include only the principal components. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section with one observation per city. Years: firm changes are measured from 2000 to 2007; controls are measured in 2000. Outcome scale: columns 2–4, 8, and 9 report level changes in firm counts; column 5 is IHS-transformed; column 6 is expressed per capita; column 7 is employment in thousands. Because these city-level models are cross-sectional with one observation per city, city-level clustering is not separately informative; province-level clustering allows arbitrary correlation in unobserved shocks across cities within the same province. The first-stage F row reports the province-clustered excluded-instrument F statistic for each column's exact sample and controls.

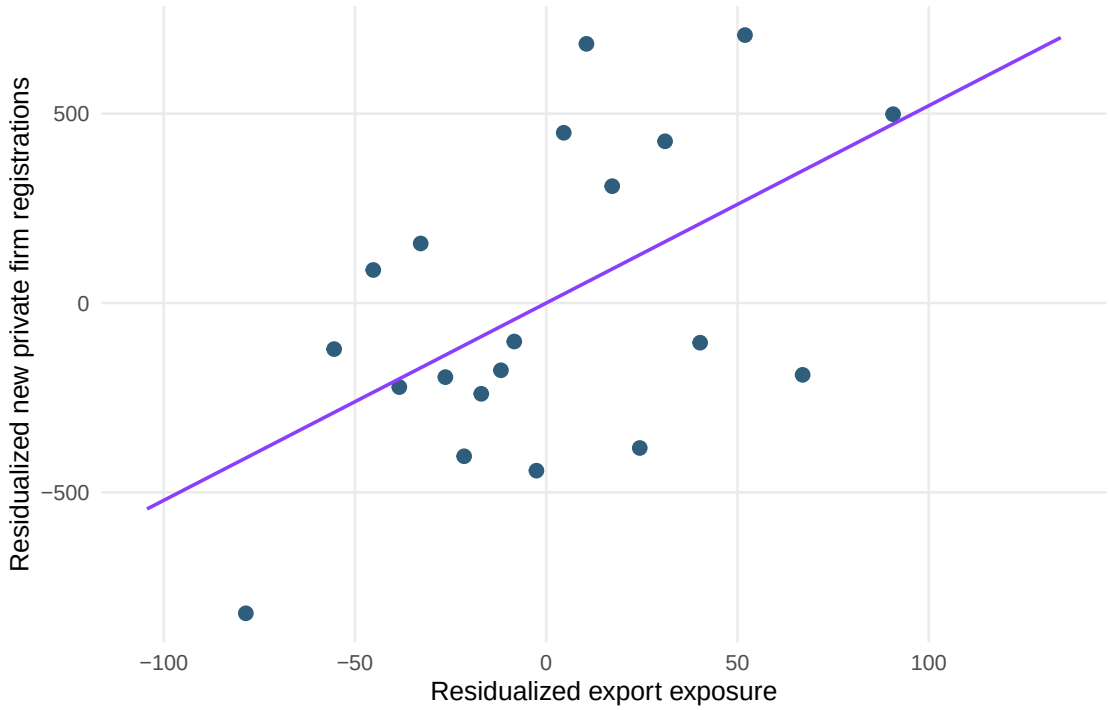


Figure 4. Export exposure and private-firm registrations

Note. Binned scatter of residualized export exposure and residualized private-firm registrations after controlling for regional fixed effects, baseline industrial principal components, and baseline census controls.

5.2 Frontline CTA Recruitment

Export exposure also shifted CTA recruitment toward frontline branch offices rather than city headquarters. Table 2 reports 2SLS estimates for CTA recruitment positions. Except for the smoothed measure in Model 4, the outcomes are measured in 2007 levels. They capture recruitment positions, not staffing stocks or first differences.

Models 1–3 progressively add controls. In Model 3, moving from the 25th to the 75th percentile of exposure implies roughly 5.9 additional CTA recruitment positions in 2007, relative to a city mean of about eight. The shift is from 74.4 to 151.9 exposure units in the included sample (Appendix Table A5). Model 4 averages openings over 2006–2008 to smooth annual volatility. The estimated effect using this measure is smaller but remains statistically significant. Models 5 and 6 locate the response in county branches. The same interquartile shift implies about six additional branch positions, with a headquarters point estimate close to zero.

Table 2. Export Exposure and CTA Recruitment Positions

	All positions, 2007			06–08 Mean	Branch	Headquarters
	(1)	(2)	(3)	(4)	(5)	(6)
Export Exposure	0.107***	0.099***	0.075***	0.037**	0.077***	−0.001
	(0.027)	(0.031)	(0.027)	(0.015)	(0.027)	(0.002)
Region FE	Y	Y	Y	Y	Y	Y
PCA Ind	N	Y	Y	Y	Y	Y
Controls	N	N	Y	Y	Y	Y
First-stage F	255.0	215.4	167.6	167.6	167.6	167.6
Num. obs.	312	312	312	312	312	312

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: tax-administration outcomes are CTA recruitment-position counts; column 4 is the 2006–2008 mean count; columns 5–6 split branch and headquarters counts. Transformations: outcomes are unlogged levels; export exposure is instrumented with tariff-predicted export exposure in 2SLS columns. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure; columns marked PCA include only the principal components. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section with one observation per city. Years: main CTA recruitment outcomes are measured in 2007, except the mean column uses 2006–2008; controls are measured in 2000. Outcome scale: all outcomes are levels; no outcome is IHS-transformed or expressed per capita. Because these city-level models are cross-sectional with one observation per city, city-level clustering is not separately informative; province-level clustering allows arbitrary correlation in unobserved shocks across cities within the same province. The first-stage F row reports the province-clustered excluded-instrument F statistic for each column's exact sample and controls.

The location of recruitment links bureaucratic expansion to tax collection. Branches administer local firms' taxes, whereas headquarters openings would indicate broader administrative expansion. The branch response and the retirement check in Appendix H.7 support the interpretation that recruitment targeted collection needs.

The firm-level analysis in Appendix G provides complementary evidence. Among firms presumed to fall under CTA collection under the 2002 default rule, reported profits moved closer to a constructed profit benchmark between 2002 and 2007, suggesting a lower level of tax avoidance. Appendix Figure A7 summarizes this reporting alignment.

5.3 EIT and Central Tax Revenue

Table 3 shows that export exposure increased fiscal-book EIT and the center's EIT share. The first two columns report the center's EIT and VAT shares, calculated from fiscal-book revenue using the legal sharing rules. Column 3 reports EIT's share of combined EIT and VAT revenue. Column 4 compares the center's EIT share with its combined VAT and consumption-tax revenue. All outcomes exclude tax-rebate transfers returned to local budgets.

The city-level EIT effect is large. Moving from the 25th to the 75th percentile of realized export exposure raises the center's EIT share by 0.395 log points, or 48.4% ($p = 0.037$). Under the 2002 rules, the CTA collected EIT from most newly established firms, and the center received 60% of EIT revenue.

VAT provides a useful comparison because, unlike EIT, its collection and sharing rules were unchanged by the 2002 reform. The CTA already administered VAT and remitted the center's 75% share before China's WTO entry. The estimated increase in the center's VAT share is smaller—15.9% across the exposure interquartile range—and statistically imprecise ($p = 0.342$). The composition results then examine whether export exposure shifted revenue toward EIT.

Columns 3 and 4 show a shift toward EIT in the tax revenue mix. An export boom could raise profits and value added together. Proportional growth would leave $EIT/(EIT + VAT)$ unchanged. Across the exposure interquartile range, however, EIT's share is 4.64 percentage points higher in 2007 ($p = 0.001$). Controlling for its 2000 value yields a 4.46-percentage-point difference ($p = 0.002$). The EIT-to-VAT ratio is about 28% higher in the 2007 specification, 22% higher with the baseline ratio controlled, and 21% higher with the 1997–2000 pre-trend also controlled. The estimates remain stable under winsorized-share, log-odds, and rank transformations.

The comparison with combined VAT and consumption-tax revenue gives a similar result. The log ratio of the center's EIT share to these revenues is 0.222 log points, or 24.8%, higher across the exposure interquartile range ($p = 0.008$). This extends the benchmark to VAT and consumption tax, whose collection and sharing rules were unchanged. Composition therefore offers a sharper test than tax levels alone. EIT gained relative to taxes outside the reform and accounted for a larger proportion of these central revenues.

The EIT gains indicate that export expansion enlarged the revenue the center could command under the post-2002 rules. Most new firms fell under CTA jurisdiction, and the center's 60% share gave it a fiscal stake in the expanding tax base. The increase in recruitment positions at frontline branches is consistent with investment in collecting this revenue. Figure 5 places these local findings in national context: CTA EIT collection diverged sharply from LTA collection after 2002.

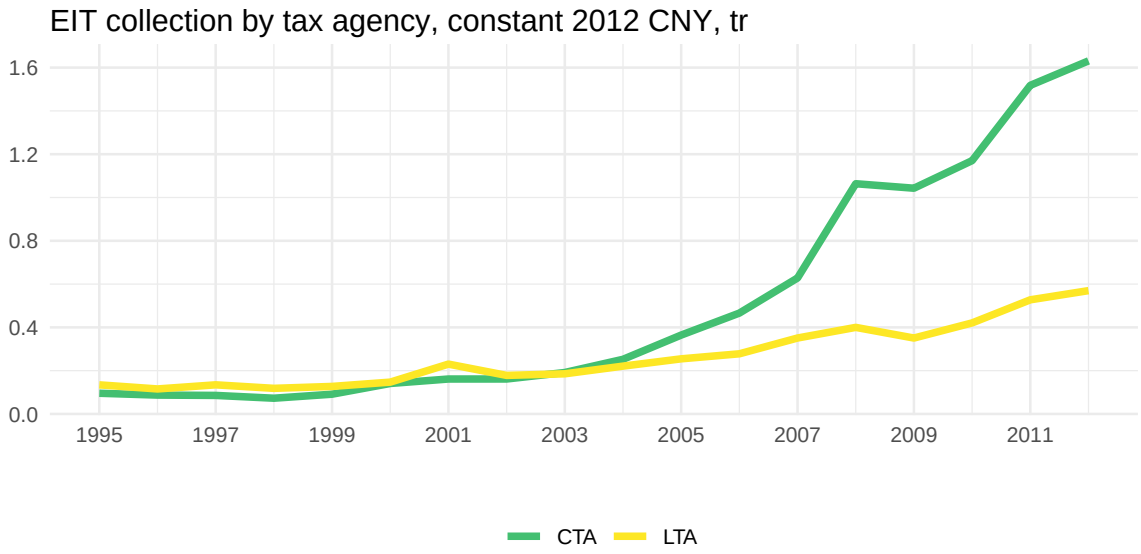


Figure 5. National EIT collection by central and local tax agencies, 1995–2012

Note. The figure reports national EIT collections by the CTA and LTA, measured in trillions of constant 2012 CNY. CTA collection diverges sharply from LTA collection after the 2002 EIT reform. Data source: *China Taxation Yearbook*, various years.

Table 3. Export Exposure and Reform-Aligned Tax Outcomes

	Central Revenue		Tax Composition	
	EIT share	VAT share	EIT / (EIT + VAT)	EIT / (VAT + consumption tax)
Export Exposure	0.005**	0.002	0.059***	0.003***
	(0.002)	(0.002)	(0.017)	(0.001)
Effect across exposure IQR	+48.4%	+15.9%	+4.64 pp	+24.8%
Num. obs.	310	310	310	309
First-stage F	139.4	139.4	139.4	139.3
Region FE	Y	Y	Y	Y
PCA industry controls	Y	Y	Y	Y
Baseline census controls	Y	Y	Y	Y

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Each column reports a city-level 2SLS regression of the 2007 outcome on realized export exposure, instrumented by tariff-predicted exposure. All specifications include regional fixed effects, the first three principal components of baseline industrial structure, and baseline census controls; standard errors are clustered by province. Columns 1 and 2 report the central government's EIT and VAT revenue shares, calculated as 1.5 times fiscal-book EIT under the 60:40 sharing rule and three times fiscal-book VAT under the 75:25 rule. These are amounts inferred from the legal sharing rules, not observed city-level CTA collections. Column 3 reports fiscal-book EIT divided by combined fiscal-book EIT and VAT, in percentage points. Column 4 reports the log ratio of the center's EIT share to the sum of the reported 75% VAT remittance and consumption-tax revenue. The outcomes in columns 1, 2, and 4 are logged. The interquartile-range row converts logged effects to percent changes and reports the share effect in percentage points.

6 Fiscal Compensation for SOE Restructuring

I examine whether localities bearing greater SOE restructuring burdens received more grants, drawing first on policy evidence and then on the city panel.

6.1 Policy Evidence

Central leaders treated SOE unemployment as a fiscal obligation and a threat to social order. In a 1998 speech, Premier Zhu Rongji warned that unresolved layoffs endangered livelihoods, reemployment, and social order, creating "instability in people's hearts and society" (R. Zhu, 1998). Limited state funds required the government to prioritize laid-off SOE workers. Central special transfers would support central and western regions and old industrial bases. Fiscal compensation thus became an explicit national priority.

The 2000 urban social-security pilot specified responsibilities for pensions, unemployment insurance, minimum-livelihood guarantees, reemployment, and financing social protection during restructuring (State Council, 2000). The center issued these directives but placed the benefit obligations on local budgets.

SOE workers' claims rested on prior public employment and welfare commitments. Their losses were concentrated in identifiable cities, where local budgets carried the resulting obligations (Y. Cai, 2002; Friedman, 2014; Hurst & O'Brien, 2002; C. K. Lee, 2007; Luo et al., 2023). Survey evidence is consistent with these demands for compensation. In the 2003 CGSS, respondents reporting SOE layoffs were more likely to support taxing the rich to help the poor and to prioritize unemployment reduction (Appendix Figure A4 and Table A2).

Fiscal compensation through grants could both finance social protection and replace revenue lost during SOE contraction. Figure 6 shows the expansion of central revenue and transfers during the 2000s (J. Jiang & Zhang, 2020; Lou & Wang, 2008; Wan et al., 2015). The city panel below examines whether fiscal compensation was greater where restructuring burdens were heavier.

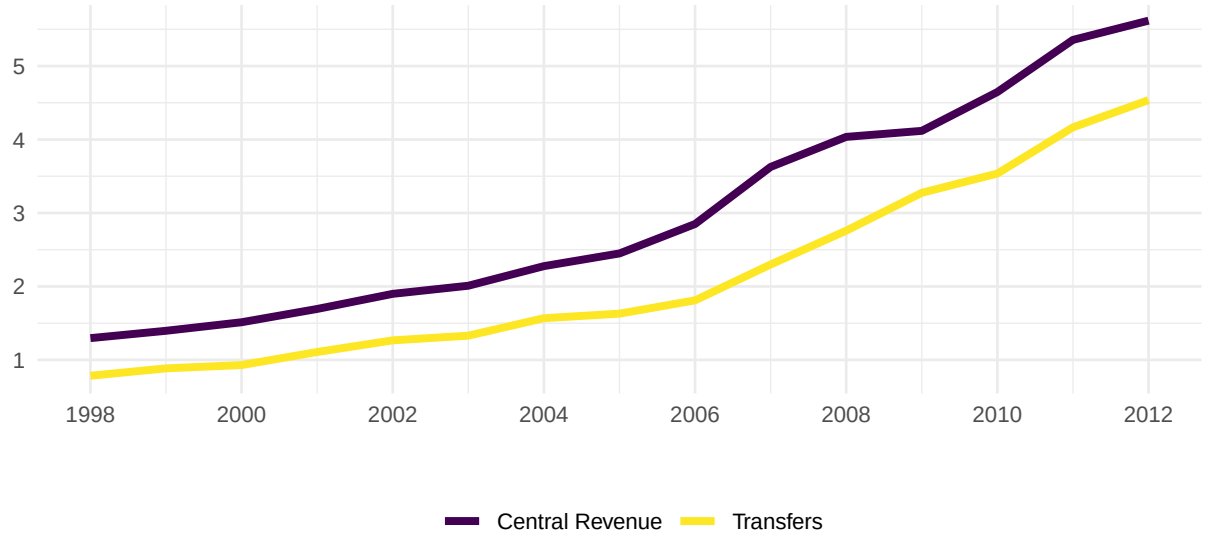


Figure 6. Expansion of central-to-local fiscal transfers

Note. The figure reports central government revenue and central-to-local fiscal transfers. Both series are expressed in trillions of Chinese yuan (CNY). Data source: *China Finance Yearbook*, various years.

6.2 Grants and Restructuring Obligations

Reported grant receipts and the number of recorded retirees increased as SOE employment shares contracted between 2000 and 2007. Table 4 estimates within-city associations with the remaining SOE employment share. All models include city and year fixed effects, with city-clustered standard errors. Models with controls add the share of the population employed in manufacturing, the ethnic-minority share, GDP per capita, the working-age share, and out-of-province migrants. Negative coefficients indicate larger outcomes where the SOE employment share declined more.

The first three columns use grants reconstructed from explicitly labeled receipt lines. Greater contraction is associated with higher total grants and greater dependence on grant finance. Appendix Table A21 reports no-control estimates for total, per-capita, and per-SOE measures. A ten-percentage-point lower SOE employment share corresponds to about 4.2% more grants in the no-control model. With controls, the same decline corresponds to about 3.7% more grant revenue per resident.

Table 4. SOE Employment Share, Reported Grant Receipts, and Restructuring-Related Outcomes

	Reported Grant Receipts			Restructuring-Related Outcomes		Fiscal Dependence
	Total	Per Cap.	Per SOE	Soc. Sec.	Recorded Retirees	Grants/Spend
SOE employment share (remaining)	-0.413^{***} (0.077)	-0.360^{***} (0.094)	-3.570^{***} (0.491)	-3.468^{***} (0.516)	-0.128^{**} (0.052)	-0.117^{*} (0.067)
City FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Controls	N	Y	Y	Y	Y	Y
Num. obs.	666	664	664	664	664	664

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: reported intergovernmental grant receipts and social-security spending use source fiscal-statistics currency units; recorded retirees are counts; fiscal dependence is a grants-to-spending log ratio. Transformations: reported grant receipts are log(y + 1); per-capita outcomes subtract log total population; per-SOE outcomes subtract log(SOE employment + 1); social-security and retiree logs are winsorized; fiscal dependence is log grants minus log total spending. Controls: columns marked Controls include the share of the population employed in manufacturing, ethnic-minority share, GDP per capita, working-age share, and log out-of-province migrants. FE: city and year fixed effects in every column. SE: standard errors clustered by city. Sample: two-period city panel. Years: panel years are 2000 and 2007. Outcome scale: columns 1, 5, and 6 are logged outcomes; column 2 reports logged grants adjusted for total population; columns 3 and 4 are per-SOE logs; no outcome is IHS-transformed. The grant measure sums fiscal-book receipt lines explicitly labeled tax rebates, general transfers, special transfers, and other subsidies from higher levels of government. Column 1 omits the added controls; columns 2–6 include them. The regressor is SOE employment share (remaining), so negative coefficients indicate larger outcomes in cities with larger SOE employment-share declines. Appendix H.6 documents schedule harmonization and component checks.

City fixed effects absorb persistent differences in industrial history, administrative status, and fiscal scale. Total grants measure the resource response, per-capita grants adjust for population, and fiscal dependence measures reliance on grant finance. The convergence of these measures is consistent with greater fiscal compensation and fiscal dependence where contraction was deeper.

Recorded retirees are people whose pensions are carried on government accounts and budgeted through local spending. In the controlled model, a ten-percentage-point decline in the SOE employment share is associated with about 1.3% more recorded retirees. Some layoffs during this period involved contract buyouts that effectively moved workers into early retirement. Appendix Table A21 also reports no-control specifications for social-security spending per remaining SOE worker and recorded retirees. Utility spending, other general spending, and public employment show no consistent response (Appendix Table A22). The pattern weighs against a uniform budget expansion and points to fiscal compensation for restructuring-related obligations and revenue pressures.

6.3 The Geography of Revenue and Fiscal Compensation

Export expansion and SOE contraction were concentrated in different cities, but China's national transfer system allowed revenue from export growth to support cities facing SOE restructuring.

The State Council's plan implementing the 2002 EIT reform explicitly connected additional central income-tax revenue to general transfers, while the city panel shows that grant receipts rose with declines in the SOE employment share.

Figure 7 compares the locations of EIT revenue growth and fiscal compensation. EIT growth is higher among cities in the top quartile of export exposure, while grant receipts and the number of recorded retirees increase more in cities in the top quartile of SOE employment-share decline. I find only partial overlap between the two top quartiles.

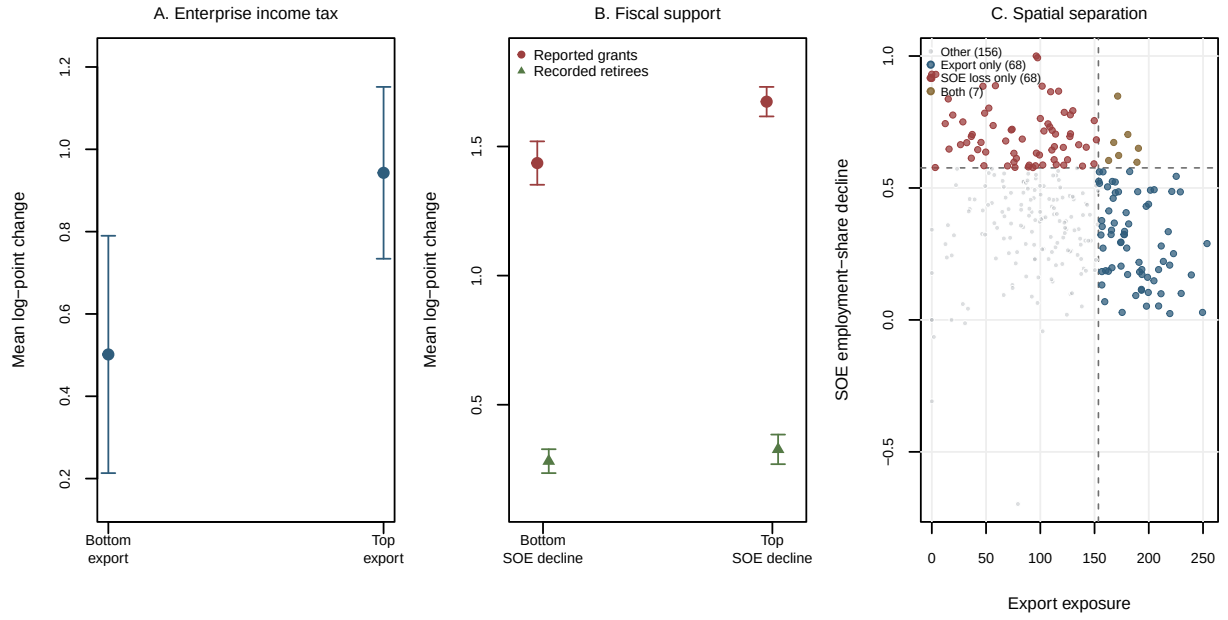


Figure 7. City-level evidence on fiscal pooling

Note. City-level descriptive evidence from the common complete-case sample. Panel A reports the mean 2000–2007 change in log fiscal-book EIT for cities in the bottom and top quartiles of export exposure. Panel B reports mean 2000–2007 changes in log reported intergovernmental grant receipts and winsorized log recorded retirees for cities in the bottom and top quartiles of SOE employment-share decline. In both panels, error bars show 95% confidence intervals around group means. Panel C plots the decline in SOE employment share against export exposure. Dashed lines mark the top-quartile cutoffs. Highlighted points identify cities in only the top export quartile, only the top SOE-decline quartile, and both top quartiles; gray points are all remaining cities.

National pooling allows cities facing SOE restructuring to finance social obligations with revenue collected elsewhere. A declining industrial city may face rising pension and unemployment obligations as its tax base shrinks. Central collection and revenue sharing make tax revenue from growing cities available for transfers to cities facing these pressures.

The export analysis identifies the effect of export exposure on firm registration, investment in fiscal capacity, and EIT revenue, while the SOE panel connects restructuring to higher grant receipts. The geographic comparison shows only partial overlap between cities with high export exposure and those with large declines in the SOE employment share. Together with the transfer rules, these findings are consistent with fiscal compensation reaching restructuring cities beyond the main centers of export growth.

7 Conclusion

Trade openness builds fiscal capacity when export expansion brings new firms into a tax system that gives the center a legal claim to revenue and control over collection. Post-WTO China illustrates this argument. Export exposure increased private-firm registrations and private-sector employment, while recruitment in the central tax bureaucracy expanded mainly in county branches responsible for collection. Under the 2002 reform, most new firms came under central collection and the center gained a share of their EIT revenue. The center's EIT share grew, and EIT accounted for a larger proportion of combined EIT and VAT revenue.

SOE restructuring, rooted in domestic reform, generated concentrated demands for fiscal compensation. In the city panel, larger declines in the SOE employment share were associated with higher grant receipts, greater reliance on grant finance, and more recorded retirees. Utility spending, other general spending, and public employment showed no comparable response. Cities in the top quartiles of export exposure and SOE employment-share decline overlapped only partially. Together with the transfer rules, this pattern is consistent with fiscal compensation reaching cities outside the main centers of export growth.

Embedded fiscal centralization enables the center to capture revenue and fund its political priorities. Registered entry raises the return to expanding the tax bureaucracy, while central control over collectors limits local bargaining over enforcement. These institutions shape who captures new revenue and where administrative capacity expands. National pooling then allows the center

to sustain locally administered benefits for politically organized groups able to exert pressure for support. Such support can also deepen recipient localities' dependence on the center.

The source of redistributive demand varies across settings. In China, SOE restructuring generated the claims examined here. Elsewhere, demands may arise from import-related job losses or environmental damage. The general argument concerns how revenue from expanding economic activity becomes available to finance the center's redistributive priorities.

The broader comparative question concerns who captures the fiscal returns from economic expansion. Collection authority, revenue sharing, and supervision of collectors shape central–local bargaining in federal and unitary systems (Oates, 2005; Rodden, 2004). China is a most-likely, scope-defining case: its registration and reporting systems, CTA network, statutory revenue claims, and transfer channels favored both central revenue capture and fiscal compensation. The argument applies where governments can record firms' activity, enforce central claims through accountable collectors, and finance locally administered obligations from pooled revenue. Political centralization may reduce veto points and facilitate targeted support.

The mechanism faces less favorable conditions today. Growth has slowed since the COVID-19 pandemic. Lower land-lease revenue and rising local debt have tightened fiscal space as social-spending needs have grown (World Bank, 2025). Trade conflict and tariff uncertainty have also redirected exports and investment and altered supply-chain relationships (Heise et al., 2024; L. Jiang et al., 2023; Zeng & Kim, 2025). These pressures make it harder for new-firm entry and tax-base growth to outpace claims on transfers. The framework predicts sharper central–local conflict when pooled resources grow more slowly than local obligations. Future comparative research can test the mechanism by examining how firms enter the tax system, who collects their taxes, and how revenue is pooled and allocated.

Notes

1. Appendix Figure A3 shows that tariff revenue remained small relative to total tax revenue during the reform period.
2. Appendix Table A1 summarizes the default EIT collection rule. Appendix Figure A5 reports observed collection jurisdiction by ownership and establishment cohort in the 2007 National Enterprise Tax Survey. Appendix Figure A6 documents the post-2002 shift toward CTA collection in national EIT series disaggregated by ownership.
3. The ASMF contains balance-sheet data for all SOE manufacturers and non-state manufacturers with annual sales above 5 million Chinese yuan (CNY). Following Fan, Lai, and Li (2015), I remove observations with missing identifiers or variables required for output, employment, ownership, city, or industry measures. I also exclude firms with fewer than ten employees, duplicate records, and accounting inconsistencies. These include fixed or liquid assets exceeding total assets, current assets exceeding total assets, and paid-in capital inconsistent with the sum of ownership components. Firm-level variables are winsorized before aggregation. The `gbcrosswalk` R package harmonizes industry codes across GB vintages and maps them to International Standard Industrial Classification (ISIC) and Harmonized System (HS) codes.
4. Appendix Table A24 shows that provincial staff age composition does not consistently predict recruitment, reducing concern that recruitment primarily reflects replacement of retiring staff.
5. The IHS transformation is $\text{IHS}(y) = \sinh^{-1}(y) = \log(y + \sqrt{y^2+1})$. It approximates a logarithmic transformation for large positive values while remaining defined at zero and for negative changes (Bellemare & Wichman, 2020; Burbidge et al., 1988).

References

Abramovsky, L., Klemm, A., & Phillips, D. (2014). Corporate tax in developing countries: Current trends and design issues. *Fiscal Studies*, *35*(4), 559–588. https://doi.org/10.1111/j.1475-5890.2014.12042.x

Adserà, A., & Boix, C. (2002). Trade, democracy, and the size of the public sector: The political underpinnings of openness. *International Organization*, *56*(2), 229–262. https://doi.org/10.1162/002081802320005478

Albertus, M. (2024). Authoritarian redistribution. In A. Wolf (Ed.), *The Oxford handbook of authoritarian politics*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198871996.013.22

Autor, D. H., Dorn, D., & Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, *103*(6), 2121–2168. https://doi.org/10.1257/aer.103.6.2121

Autor, D. H., Dorn, D., & Hanson, G. H. (2016). The China shock: Learning from labor-market adjustment to large changes in trade. *Annual Review of Economics*, *8*(1), 205–240. https://doi.org/10.1146/annurev-economics-080315-015041

Autor, D. H., Dorn, D., Hanson, G. H., & Majlesi, K. (2020). Importing political polarization? The electoral consequences of rising trade exposure. *American Economic Review*, *110*(10), 3139–3183. https://doi.org/10.1257/aer.20170011

Avi-Yonah, R. S. (2000). Globalization, tax competition, and the fiscal crisis of the welfare state. *Harvard Law Review*, *113*(7), 1573–1676. https://doi.org/10.2307/1342445

Baccini, L., & Weymouth, S. (2021). Gone for good: Deindustrialization, White voter backlash, and US presidential voting. *American Political Science Review*, *115*(2), 550–567. https://doi.org/10.1017/S0003055421000022

Bachas, P., Fisher-Post, M. H., Jensen, A., & Zucman, G. (2022, March). *Capital taxation, development, and globalization: Evidence from a macro-historical database* (NBER Working

Paper No. 29819). National Bureau of Economic Research. https://doi.org/10.3386/w29819

Bartik, T. J. (1991). *Who benefits from state and local economic development policies?* W.E. Upjohn Institute for Employment Research. https://doi.org/10.17848/9780585223940

Basri, M. C., Felix, M., Hanna, R., & Olken, B. A. (2021). Tax administration versus tax rates: Evidence from corporate taxation in Indonesia. *American Economic Review*, *111*(12), 3827–3871. https://doi.org/10.1257/aer.20201237

Bastiaens, I., & Rudra, N. (2018). *Democracies in peril: Taxation and redistribution in globalizing economies*. Cambridge University Press. https://doi.org/10.1017/9781108556668

Baunsgaard, T., & Keen, M. (2010). Tax revenue and (or?) trade liberalization. *Journal of Public Economics*, *94*(9–10), 563–577. https://doi.org/10.1016/j.jpubeco.2009.11.007

Bellemare, M. F., & Wichman, C. J. (2020). Elasticities and the inverse hyperbolic sine transformation. *Oxford Bulletin of Economics and Statistics*, *82*(1), 50–61. https://doi.org/10.1111/obes.12325

Beramendi, P., & Rogers, M. (2020). Fiscal decentralization and the distributive incidence of the Great Recession. *Regional Studies*, *54*(7), 881–896. https://doi.org/10.1080/00343404.2019.1652895

Besley, T., & Persson, T. (2009). The origins of state capacity: Property rights, taxation, and politics. *American Economic Review*, *99*(4), 1218–1244. https://doi.org/10.1257/aer.99.4.1218

Blanchard, O., & Shleifer, A. (2001). Federalism with and without political centralization: China versus Russia. *IMF Staff Papers*, *48*, 171–179. https://doi.org/10.2307/4621694

Boadway, R., & Sato, M. (2009). Optimal tax design and enforcement with an informal sector. *American Economic Journal: Economic Policy*, *1*(1), 1–27. https://doi.org/10.1257/pol.1.1.1

Bombardini, M., & Li, B. (2020). Trade, pollution and mortality in China. *Journal of International Economics*, *125*, 103321. https://doi.org/10.1016/j.jinteco.2020.103321

Boone, C. (1994). Trade, taxes, and tribute: Market liberalizations and the new importers in West Africa. *World Development*, *22*(3), 453–467. https://doi.org/10.1016/0305-750X(94)90135-X

Borusyak, K., Hull, P., & Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of Economic Studies*, *89*(1), 181–213. https://doi.org/10.1093/restud/rdab030

Brandt, L., Van Biesebroeck, J., Wang, L., & Zhang, Y. (2017). WTO accession and performance of Chinese manufacturing firms. *American Economic Review*, *107*(9), 2784–2820. https://doi.org/10.1257/aer.20121266

Bräutigam, D., Fjeldstad, O.-H., & Moore, M. (Eds.). (2008). *Taxation and state-building in developing countries: Capacity and consent*. Cambridge University Press. https://doi.org/10.1017/CBO9780511490897

Broz, J. L., Frieden, J., & Weymouth, S. (2021). Populism in place: The economic geography of the globalization backlash. *International Organization*, *75*(2), 464–494. https://doi.org/10.1017/S0020818320000314

Buettner, T., & Madzharova, B. (2018). WTO membership and the shift to consumption taxes. *World Development*, *108*, 197–218. https://doi.org/10.1016/j.worlddev.2018.02.020

Burbidge, J. B., Magee, L., & Robb, A. L. (1988). Alternative transformations to handle extreme values of the dependent variable. *Journal of the American Statistical Association*, *83*(401), 123–127. https://doi.org/10.1080/01621459.1988.10478575

Cagé, J., & Gadenne, L. (2018). Tax revenues and the fiscal cost of trade liberalization, 1792–2006. *Explorations in Economic History*, *70*, 1–24. https://doi.org/10.1016/j.eeh.2018.07.004

Cai, H., & Liu, Q. (2009). Competition and corporate tax avoidance: Evidence from Chinese industrial firms. *The Economic Journal*, *119*(537), 764–795. https://doi.org/10.1111/j.1468-0297.2009.02217.x

Cai, Y. (2002). The resistance of Chinese laid-off workers in the reform period. *The China Quarterly*, *170*, 327–344. https://doi.org/10.1017/S0009443902000219

Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, *90*(3), 414–427. https://doi.org/10.1162/rest.90.3.414

Cameron, A. C., & Miller, D. L. (2015). A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, *50*(2), 317–372. https://doi.org/10.3368/jhr.50.2.317

Cameron, D. R. (1978). The expansion of the public economy: A comparative analysis. *American Political Science Review*, *72*(4), 1243–1261. https://doi.org/10.2307/1954537

Campello, D., & Urdinez, F. (2021). Voter and legislator responses to localized trade shocks from China in Brazil. *Comparative Political Studies*, *54*(7), 1131–1162. https://doi.org/10.1177/0010414020970233

Cantoni, D., Mohr, C., & Weigand, M. (2024). The rise of fiscal capacity: Administration and state consolidation in the Holy Roman Empire. *Econometrica*, *92*(5), 1439–1472. https://doi.org/10.3982/ECTA20612

Chaney, T. (2008). Distorted gravity: The intensive and extensive margins of international trade. *American Economic Review*, *98*(4), 1707–1721. https://doi.org/10.1257/aer.98.4.1707

Chen, J., Pan, J., & Xu, Y. (2016). Sources of authoritarian responsiveness: A field experiment in China. *American Journal of Political Science*, *60*(2), 383–400. https://doi.org/10.1111/ajps.12207

Choi, E. K. (2016). The politics of central tax collection in China since 1994: Local collusion and political control. *Journal of Contemporary China*, *25*(97), 146–159. https://doi.org/10.1080/10670564.2015.1060770

Colantone, I., & Stanig, P. (2018). The trade origins of economic nationalism: Import competition and voting behavior in Western Europe. *American Journal of Political Science*, *62*(4), 936–953. https://doi.org/10.1111/ajps.12358

Desai, M. A., & Dharmapala, D. (2006). Corporate tax avoidance and high-powered incentives. *Journal of Financial Economics*, *79*(1), 145–179. https://doi.org/10.1016/j.jfineco.2005.02.002

Desai, R. M., Olofsgård, A., & Yousef, T. M. (2009). The logic of authoritarian bargains. *Economics & Politics*, *21*(1), 93–125. https://doi.org/10.1111/j.1468-0343.2008.00337.x

Devereux, M. P., Griffith, R., & Klemm, A. (2002). Corporate income tax reforms and international tax competition. *Economic Policy*, *17*(35), 449–495. https://doi.org/10.1111/1468-0327.00094

Dincecco, M. (2009). Fiscal centralization, limited government, and public revenues in Europe, 1650–1913. *The Journal of Economic History*, *69*(1), 48–103. https://doi.org/10.1017/S0022050709000345

Eaton, K., & Dickovick, J. T. (2004). The politics of re-centralization in Argentina and Brazil. *Latin American Research Review*, *39*(1), 90–122. https://doi.org/10.1353/lar.2004.0007

Falleti, T. G. (2010). *Decentralization and subnational politics in Latin America*. Cambridge University Press. https://doi.org/10.1017/CBO9780511777813

Fan, H., Lai, E. L.-C., & Li, Y. A. (2015). Credit constraints, quality, and export prices: Theory and evidence from China. *Journal of Comparative Economics*, *43*(2), 390–416. https://doi.org/10.1016/j.jce.2015.02.007

Fan, H., Li, Y. A., & Yeaple, S. R. (2015). Trade liberalization, quality, and export prices. *The Review of Economics and Statistics*, *97*(5), 1033–1051. https://doi.org/10.1162/REST_a_00524

Farrell, H., & Newman, A. (2023). *Underground empire: How America weaponized the world economy*. Henry Holt and Company.

Frazier, M. W. (2004). China's pension reform and its discontents. *The China Journal*, *51*, 97–114. https://doi.org/10.2307/3182148

Friedman, E. (2014). *Insurgency trap: Labor politics in postsocialist China*. Cornell University Press.

Gallagher, M. E. (2005). *Contagious capitalism: Globalization and the politics of labor in China*. Princeton University Press. https://www.jstor.org/stable/j.ctt7pgn7

Garfias, F., & Sellars, E. A. (2022). When state building backfires: Elite coordination and popular grievance in rebellion. *American Journal of Political Science*, *66*(4), 977–992. https://doi.org/10.1111/ajps.12611

Garfias, F., & Sellars, E. A. (2026). Fiscal legibility and state development: Theory and evidence from colonial Mexico. *American Journal of Political Science*, *70*(1), 5–20. https://doi.org/10.1111/ajps.12901

Gnangnon, S. K., & Brun, J.-F. (2019). Trade openness, tax reform and tax revenue in developing countries. *The World Economy*, *42*(12), 3515–3536. https://doi.org/10.1111/twec.12858

Goldberg, P. K. (2023). *The unequal effects of globalization*. MIT Press.

Goldsmith-Pinkham, P., Sorkin, I., & Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, *110*(8), 2586–2624. https://doi.org/10.1257/aer.20181047

Guimbard, H., Jean, S., Mimouni, M., & Pichot, X. (2012). MAcMap-HS6 2007, an exhaustive and consistent measure of applied protection in 2007. *International Economics*, *130*, 99–121. https://doi.org/10.1016/S2110-7017(13)60046-3

Haque, M. E., & Mukherjee, A. (2005). *Free entry and government revenue under trade liberalization* (University of Nottingham Research Paper No. 2004/38). University of Nottingham. https://doi.org/10.2139/ssrn.764305

Heise, S., Pierce, J. R., Schaur, G., & Schott, P. K. (2024, February). *Tariff rate uncertainty and the structure of supply chains* (NBER Working Paper No. 32138). National Bureau of Economic Research. https://doi.org/10.3386/w32138

Hiscox, M. J. (2001). Class versus industry cleavages: Inter-industry factor mobility and the politics of trade. *International Organization*, *55*(1), 1–46. https://doi.org/10.1162/002081801551405

Holland, A. C. (2016). Forbearance. *American Political Science Review*, *110*(2), 232–246. https://doi.org/10.1017/S0003055416000083

Huang, X. (2020). *Social protection under authoritarianism: Health politics and policy in China*. Oxford University Press.

Huang, Y. (1996). Central-local relations in China during the reform era: The economic and institutional dimensions. *World Development*, *24*(4), 655–672. https://doi.org/10.1016/0305-750X(95)00160-E

Hurst, W. (2009). *The Chinese worker after socialism*. Cambridge University Press.

Hurst, W., & O'Brien, K. J. (2002). China's contentious pensioners. *The China Quarterly*, *170*, 345–360. https://doi.org/10.1017/S0009443902000220

Jacoby, H. G., Mansuri, G., & Fatima, F. (2021). Decentralizing corruption: Irrigation reform in Pakistan. *Journal of Public Economics*, *202*, 104499. https://doi.org/10.1016/j.jpubeco.2021.104499

Jia, R., & Nie, H. (2017). Decentralization, collusion, and coal mine deaths. *The Review of Economics and Statistics*, *99*(1), 105–118. https://doi.org/10.1162/REST_a_00563

Jiang, J., & Zhang, M. (2020). Friends with benefits: Patronage networks and distributive politics in China. *Journal of Public Economics*, *184*, 104143. https://doi.org/10.1016/j.jpubeco.2020.104143

Jiang, L., Lu, Y., Song, H., & Zhang, G. (2023). Responses of exporters to trade protectionism: Inferences from the US-China trade war. *Journal of International Economics*, *140*, 103687. https://doi.org/10.1016/j.jinteco.2022.103687

Keen, M., & Ligthart, J. E. (2002). Coordinating tariff reduction and domestic tax reform. *Journal of International Economics*, *56*(2), 489–507. https://doi.org/10.1016/S0022-1996(01)00123-4

Keen, M., & Ligthart, J. E. (2005). Coordinating tariff reduction and domestic tax reform under imperfect competition. *Review of International Economics*, *13*(2), 385–390. https://doi.org/10.1111/j.1467-9396.2005.00510.x

Keen, M., & Lockwood, B. (2010). The value added tax: Its causes and consequences. *Journal of Development Economics*, *92*(2), 138–151. https://doi.org/10.1016/j.jdeveco.2009.01.012

Khattry, B., & Rao, J. M. (2002). Fiscal faux pas?: An analysis of the revenue implications of trade liberalization. *World Development*, *30*(8), 1431–1444. https://doi.org/10.1016/S0305-750X(02)00043-8

Kleven, H. J., Kreiner, C. T., & Saez, E. (2016). Why can modern governments tax so much? An agency model of firms as fiscal intermediaries. *Economica*, *83*(330), 219–246. https://doi.org/10.1111/ecca.12182

Kong, D., & Liu, C. (2024). Centralization and regulatory enforcement: Evidence from personnel authority reform in China. *Journal of Public Economics*, *229*, 105030. https://doi.org/10.1016/j.jpubeco.2023.105030

Kovak, B. K. (2013). Regional effects of trade reform: What is the correct measure of liberalization? *American Economic Review*, *103*(5), 1960–1976. https://doi.org/10.1257/aer.103.5.1960

Lee, C. K. (2007). *Against the law: Labor protests in China's Rustbelt and Sunbelt*. University of California Press. https://doi.org/10.1525/9780520940642

Lee, H. Y. (2000). Xiagang, the Chinese style of laying off workers. *Asian Survey*, *40*(6), 914–937. https://doi.org/10.1525/as.2000.40.6.01p0110k

Levi, M. (1988). *Of rule and revenue*. University of California Press.

Li, B. (2018). Export expansion, skill acquisition and industry specialization: Evidence from China. *Journal of International Economics*, *114*, 346–361. https://doi.org/10.1016/j.jinteco.2018.07.009

Li, H., Meng, L., Wang, Q., & Zhou, L.-A. (2008). Political connections, financing and firm performance: Evidence from Chinese private firms. *Journal of Development Economics*, *87*(2), 283–299. https://doi.org/10.1016/j.jdeveco.2007.03.001

Li, P., & Zhang, Y. (2004). The professional reintegration of the "Xiagang": A survey in Liaoning province underscores the importance of vocational training courses. *China Perspectives*, (52). https://doi.org/10.4000/chinaperspectives.802

Liu, Y. (2018). Government extraction and firm size: Local officials' responses to fiscal distress in China. *Journal of Comparative Economics*, *46*(4), 1310–1331. https://doi.org/10.1016/j.jce.2018.09.002

Lou, J. (2008). The reform of intergovernmental fiscal relations in China: Lessons learned. In J. Lou & S. Wang (Eds.), *Public finance in China: Reform and growth for a harmonious society* (pp. 155–169). World Bank. https://doi.org/10.1596/978-0-8213-6927-2

Lou, J., & Wang, S. (Eds.). (2008). *Public finance in China: Reform and growth for a harmonious society*. World Bank. https://doi.org/10.1596/978-0-8213-6927-2

Lü, X., & Landry, P. F. (2014). Show me the money: Interjurisdiction political competition and fiscal extraction in China. *American Political Science Review*, *108*(3), 706–722. https://doi.org/10.1017/S0003055414000252

Luo, Y., Wang, Y., Hong, C., He, P., & Zheng, X. (2023). The shattered "Iron Rice Bowl": Effects of Chinese state-owned enterprise reform on depressive symptoms in later life. *Social Psychiatry and Psychiatric Epidemiology*, *58*(8), 1193–1200. https://doi.org/10.1007/s00127-023-02437-9

MacKinnon, J. G., Nielsen, M. Ø., & Webb, M. D. (2023). Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, *232*(2), 272–299. https://doi.org/10.1016/j.jeconom.2022.04.001

Malesky, E. J., Nguyen, C. V., & Tran, A. (2014). The impact of recentralization on public services: A difference-in-differences analysis of the abolition of elected councils in Vietnam. *American Political Science Review*, *108*(1), 144–168. https://doi.org/10.1017/S0003055413000580

Manzon, G. B., Jr., & Plesko, G. A. (2002). The relation between financial and tax reporting measures of income. *Tax Law Review*, *55*(2), 175–214.

Mazzuca, S. (2021). *Latecomer state formation: Political geography and capacity failure in Latin America*. Yale University Press.

Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, *71*(6), 1695–1725. https://doi.org/10.1111/1468-0262.00467

Moore, M. (2015). Taxation and development. In C. Lancaster & N. van de Walle (Eds.), *The Oxford handbook of the politics of development* (pp. 224–255). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199845156.013.39

Mujumdar, S. (2004). Revenue implications of trade liberalization under imperfect competition. *Economics Letters*, *82*(1), 83–89. https://doi.org/10.1016/j.econlet.2003.07.005

Naito, T. (2006). Tariff and tax reform: Dynamic implications. *Journal of International Economics*, *68*(2), 504–517. https://doi.org/10.1016/j.jinteco.2005.04.002

Naughton, B. J. (2018). *The Chinese economy: Adaptation and growth* (2nd ed.). MIT Press.

Nooruddin, I., & Rudra, N. (2014). Are developing countries really defying the embedded liberalism compact? *World Politics*, *66*(4), 603–640. https://doi.org/10.1017/S0043887114000203

Oates, W. E. (2005). Toward a second-generation theory of fiscal federalism. *International Tax and Public Finance*, *12*(4), 349–373. https://doi.org/10.1007/s10797-005-1619-9

O'Brien, R., Schechtl, M., & Parolin, Z. (2025). Fiscal centralization and inequality in children's economic mobility. *American Sociological Review*, *90*(1), 114–141. https://doi.org/10.1177/00031224241303459

O'Donnell, G. (1993). On the state, democratization and some conceptual problems: A Latin American view with glances at some postcommunist countries. *World Development*, *21*(8), 1355–1369. https://doi.org/10.1016/0305-750X(93)90048-E

Okunogbe, O., & Tourek, G. (2024). How can lower-income countries collect more taxes? The role of technology, tax agents, and politics. *Journal of Economic Perspectives*, *38*(1), 81–106. https://doi.org/10.1257/jep.38.1.81

Pan, J. (2020). *Welfare for autocrats: How social assistance in China cares for its rulers*. Oxford University Press.

Pomeranz, D. (2015). No taxation without information: Deterrence and self-enforcement in the value added tax. *American Economic Review*, *105*(8), 2539–2569. https://doi.org/10.1257/aer.20130393

Rettl, P. (2025, September). *Turning away from the state: Trade shocks and informal insurance in Brazil* (Harvard Business School BGIE Unit Working Paper No. 25-038). Harvard Business School. https://doi.org/10.2139/ssrn.5154196

Rodden, J. (2002). The dilemma of fiscal federalism: Grants and fiscal performance around the world. *American Journal of Political Science*, *46*(3), 670–687. https://doi.org/10.2307/3088407

Rodden, J. (2004). Comparative federalism and decentralization: On meaning and measurement. *Comparative Politics*, *36*(4), 481–500. https://doi.org/10.2307/4150172

Rodrik, D. (1998). Why do more open economies have bigger governments? *Journal of Political Economy*, *106*(5), 997–1032. https://doi.org/10.1086/250038

Rodrik, D. (2000). What drives public employment in developing countries? *Review of Development Economics*, *4*(3), 229–243. https://doi.org/10.1111/1467-9361.00091

Ruggie, J. G. (1982). International regimes, transactions, and change: Embedded liberalism in the postwar economic order. *International Organization*, *36*(2), 379–415. https://doi.org/10.1017/S0020818300018993

Scheve, K., & Serlin, T. (2023). The German trade shock and the rise of the neo-welfare state in early twentieth-century Britain. *American Political Science Review*, *117*(2), 557–574. https://doi.org/10.1017/S0003055422000673

Seelkopf, L., Lierse, H., & Schmitt, C. (2016). Trade liberalization and the global expansion of modern taxes. *Review of International Political Economy*, *23*(2), 208–231. https://doi.org/10.1080/09692290.2015.1125937

Slemrod, J. (2004). Are corporate tax rates, or countries, converging? *Journal of Public Economics*, *88*(6), 1169–1186. https://doi.org/10.1016/S0047-2727(03)00061-6

Slemrod, J. (2019). Tax compliance and enforcement. *Journal of Economic Literature*, *57*(4), 904–954. https://doi.org/10.1257/jel.20181437

Solinger, D. J. (2002). Labour market reform and the plight of the laid-off proletariat. *The China Quarterly*, *170*, 304–326. https://doi.org/10.1017/S0009443902000207

Sørensen, P. B. (2007). Can capital income taxes survive? And should they? *CESifo Economic Studies*, *53*(2), 172–228. https://doi.org/10.1093/cesifo/ifm010

State Administration of Taxation. (2000, November 9). *Guojia shuiwu zongju guanyu tuixing zengzhishui fangwei shuikong xitong ruogan wenti de tongzhi [Circular on implementing the VAT anti-counterfeiting tax-control system]*. Retrieved September 26, 2026, from https://fgk.chinatax.gov.cn/zcfgk/c100012/c5193394/content.html

State Administration of Taxation. (2002, January 16). *Guanyu suodeshui shouru fenxiang tizhi gaige hou shuishou zhengguan fanwei de tongzhi [Notice on the scope of tax collection and administration after the income tax revenue-sharing system reform]*. http://www.chinatax.gov.cn/chinatax/n810341/n810765/n812203/200201/c1209868/content.html

State Council. (2000, December 25). *Guowuyuan guanyu yinfa wanshan chengzhen shehui baozhang tixi shidian fang'an de tongzhi [Circular on the urban social security system pilot plan]*. http://www.gov.cn/zhengce/content/2010-11/12/content_7429.htm

State Council. (2001, December 31). *Guowuyuan guanyu yinfa suodeshui shouru fenxiang gaige fang'an de tongzhi [Circular issuing the income tax revenue-sharing reform plan]*. https://www.gov.cn/gongbao/content/2002/content_61880.htm

Taliercio, R. R. (2004). Administrative reform as credible commitment: The impact of autonomy on revenue authority performance in Latin America. *World Development*, *32*(2), 213–232. https://doi.org/10.1016/j.worlddev.2003.08.008

Tian, B. (2014). *Zhengqi hemou, qiye taoshui yu qiye jixiao [Government-enterprise collusion, corporate tax evasion, and corporate performance]* [Unpublished doctoral dissertation]. Huazhong University of Science and Technology.

Tian, Y. (2024). International trade liberalization and domestic institutional reform: Effects of WTO accession on Chinese internal migration policy. *The Review of Economics and Statistics*, *106*(3), 794–813. https://doi.org/10.1162/rest_a_01175

Tilly, C. (1992). *Coercion, capital, and European states, AD 990–1992*. Blackwell.

Tsui, K.-y. (2005). Local tax system, intergovernmental transfers and China's local fiscal disparities. *Journal of Comparative Economics*, *33*(1), 173–196. https://doi.org/10.1016/j.jce.2004.11.003

Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review*, *108*(8), 2015–2047. https://doi.org/10.1257/aer.20141745

Véjares, M. (2025). Varieties of state-building: Ecology, clientelism, and bureaucratic rule in Chile. *Perspectives on Politics*, *23*(2), 627–648. https://doi.org/10.1017/S1537592724000033

Wan, X., Ma, Y., & Zhang, K. (2015). Political determinants of intergovernmental transfers in a regionally decentralized authoritarian regime: Evidence from China. *Applied Economics*, *47*(27), 2803–2820. https://doi.org/10.1080/00036846.2015.1008768

Wong, L., & Ngok, K. (2006). Social policy between plan and market: Xiagang (off-duty employment) and the policy of the re-employment service centres in China. *Social Policy & Administration*, *40*(2), 158–173. https://doi.org/10.1111/j.1467-9515.2006.00482.x

World Bank. (2025, December). *Advancing reforms, enhancing prospects: China economic update, December 2025*. https://thedocs.worldbank.org/en/doc/600cd53e2bb24d516b8c3489e5d2c187-0070012025/original/CEU-December-2025-EN.pdf

Yang, D. L. (2004). *Remaking the Chinese Leviathan: Market transition and the politics of governance in China*. Stanford University Press. https://doi.org/10.1515/9781503619449

Zeng, K., & Kim, S. Y. (2025). Chinese firms in the US–China trade war: Decoupling through reshoring? *International Studies Quarterly*, *69*(1), sqae117. https://doi.org/10.1093/isq/sqae117

Zhang, H., Lu, S., & Chen, S. (2024). Does centralization of tax administration regulate tax competition? Evidence from a quasi-natural experiment in China. *Economic Analysis and Policy*, *84*, 1084–1098. https://doi.org/10.1016/j.eap.2024.10.015

Zhu, H. (2025). Contentious origins of authoritarian social protection: China's "threat-driven" strategy in redistribution. *Studies in Comparative International Development*, *60*(1), 192–223. https://doi.org/10.1007/s12116-024-09429-z

Zhu, R. (1998, May 16). *Zai guoyou qiye xiagang zhigong jiben shenghuo baozhang he zai jiuye gongzuo huiyi shang de jianghua [Speech at the conference on basic living guarantees and reemployment for laid-off workers from state-owned enterprises]*. http://www.reformdata.org/1998/0516/10000.shtml

Appendix Catalog

This appendix collects institutional details, supplementary descriptive evidence, measurement procedures, and robustness checks that support the main analysis.

Appendix	Contents
Appendix A	EIT collection jurisdiction under the 2002 EIT reform.
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Appendix E	EIT collection by tax agency, ownership type, and establishment cohort.
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Appendix G	Firm-level measurement of profit-reporting alignment.
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Appendix A EIT Collection Jurisdiction under the 2002 EIT Reform

Table A1. EIT Collection Jurisdiction under the 2002 Reform

Est. Year	< 2002	≥ 2002
FIE	CTA	CTA
PE	LTA	CTA (default)
SOE	LTA	CTA (default)

This table summarizes the default EIT collection assignment by firm ownership and year of establishment. Exceptions to the default rule remained in effect. Appendix Figure A5 reports the observed 2007 distribution across agencies. The nationwide rule nevertheless shifted most newly established PEs and SOEs toward CTA administration.

Appendix B Export Expansion, Firm Registration, and Aggregate EIT Revenue

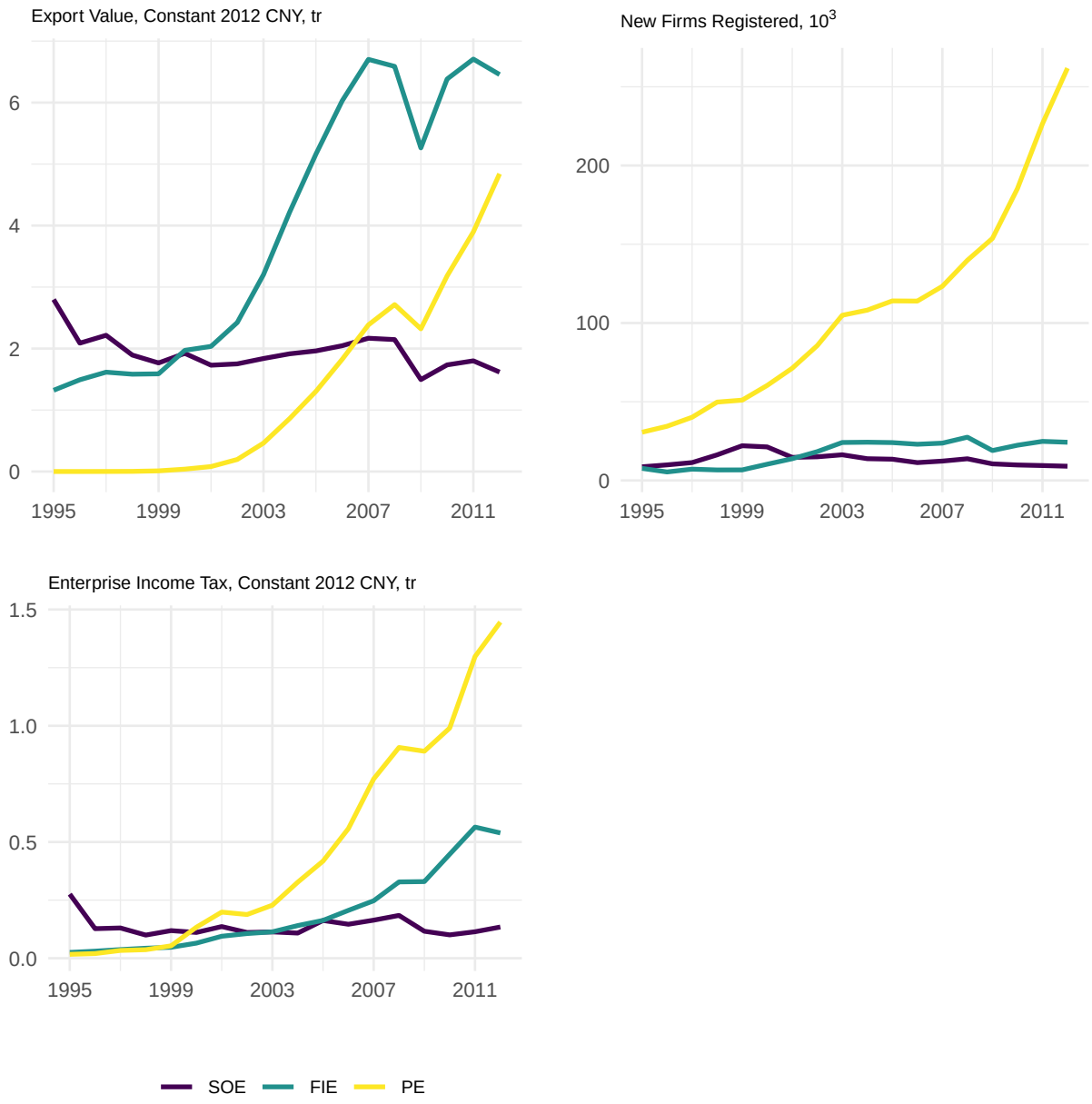


Figure A1. Export expansion, firm registration, and EIT revenue, 1995–2012

Note. The figure reports exports, new registrations, and EIT revenue by ownership type. Data sources: *China Customs Statistics Yearbook*, firm registration records, and *China Taxation Yearbook*.

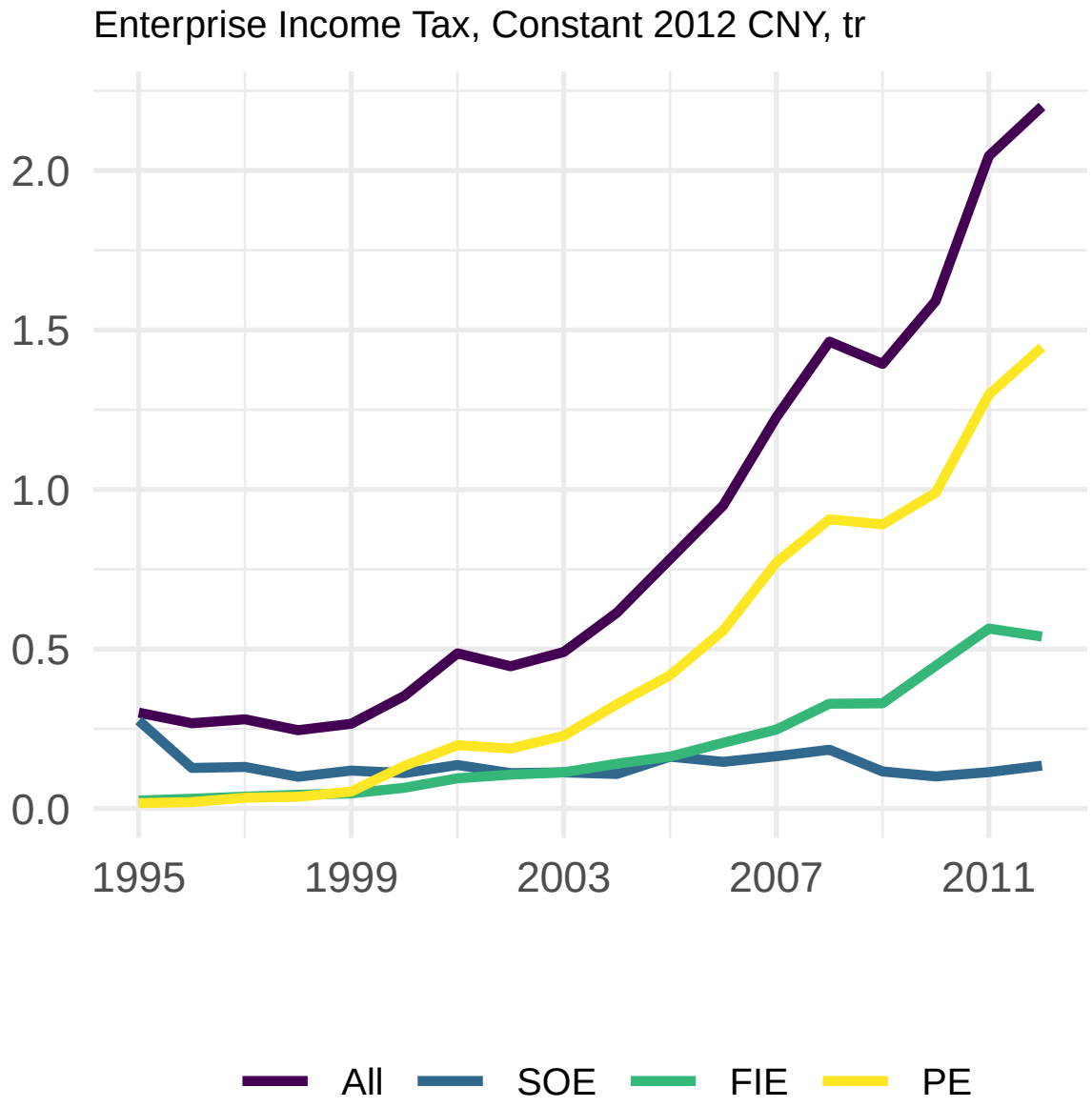


Figure A2. Total and ownership-specific EIT revenue, China, 1995–2012

Note. This figure illustrates total and ownership-specific values of EIT revenue. The difference between total EIT revenue and the sum of EIT revenue from SOEs, FIEs, and PEs is due to other ownership categories not plotted. The PE category includes proprietorships, limited liability companies, and corporations. Data source: *China Taxation Yearbook*, various years.

Appendix C Total Tax Revenue and Tariff Revenue

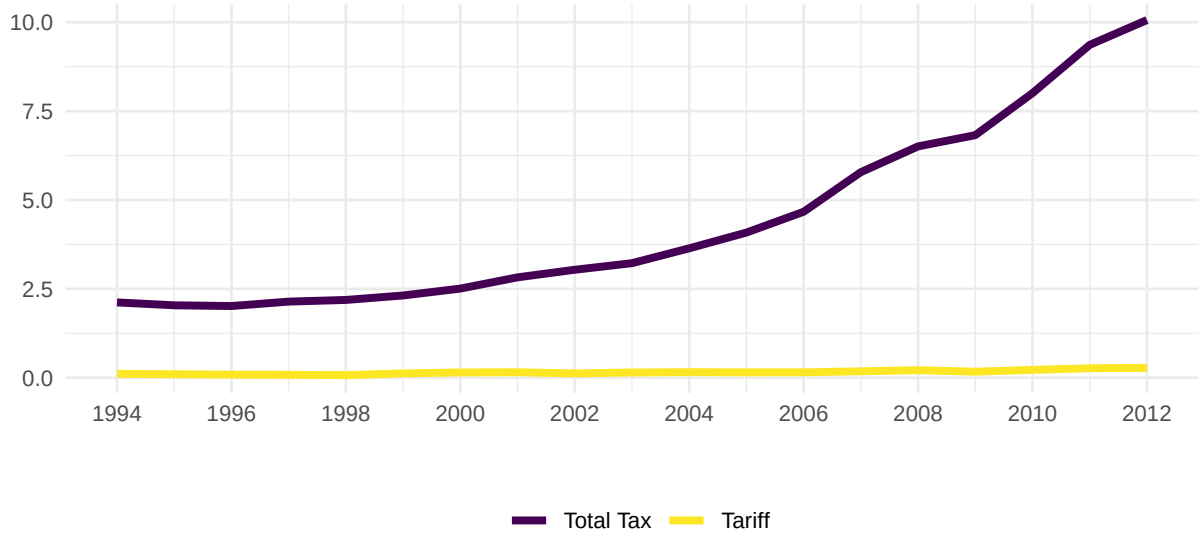


Figure A3. Total tax revenue and tariff revenue, China, 1994–2012

Note. The figure reports total tax revenue and tariff revenue in trillions of constant 2012 CNY. Data source: *China Finance Yearbook*, various years.

Appendix D Worker Attitudes Toward Redistribution

I use the 2003 CGSS, conducted by Renmin University, to describe attitudes among respondents reporting public-sector layoffs. The survey records employment histories and political and economic attitudes. I code *Job Loss* as 1 when a respondent reports having been laid off from a public-sector position in an SOE.

Figure A4 shows conditional associations between reported public-sector layoffs and key attitudinal outcomes. Full regression results are reported in Table A2, and Table A3 provides English translations of the survey questions for each outcome variable.

Conditional on the included covariates, respondents reporting an SOE layoff exhibit stronger redistributive preferences and economic grievances. They are more likely to support taxing the rich to help the poor (*Tax the Rich*) and to identify unemployment as a priority problem

(*Unemployment Priority*). They are also more likely to report a decline in perceived socioeconomic status over the past three years (*Decline in Socioeconomic Status*) and to judge their living conditions worse than those of the average citizen (*Life Worse*).

These respondents are also more likely to agree that workers deserve higher pay (*Workers Deserve High Pay*), and their estimated probability of not being employed full time is about 42 percentage points higher (*Not Employed Full Time*).

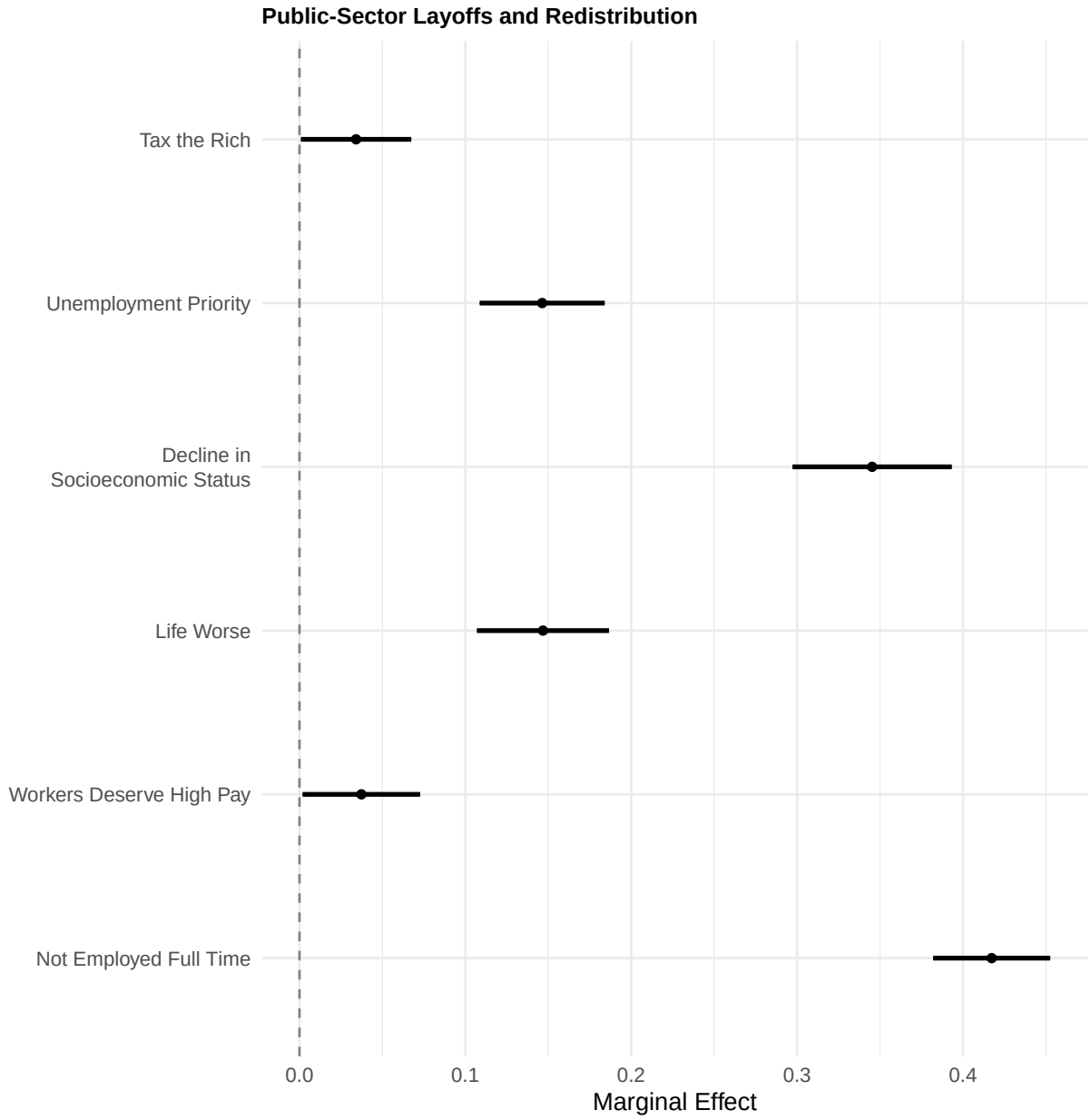


Figure A4. Workers' attitudes toward redistribution, 2003

Note. Error bars indicate 95% confidence intervals. Standard errors are robust and clustered at the township level. Regression specifications are identical to those presented in Table A2.

Table A2. Public-Sector Layoffs and Redistributive Attitudes, 2003

	Tax the Rich	Unemployment Priority	Decline in Socioeconomic Status	Life Worse	Workers Deserve High Pay	Not Employed Full Time
Job Loss	0.034**	0.146***	0.345***	0.147***	0.037**	0.417***
	(0.017)	(0.019)	(0.024)	(0.020)	(0.018)	(0.018)
Age	0.004***	−0.001	0.007***	−0.000	0.000	0.012***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Gender	−0.015	−0.051***	0.010	0.028**	−0.030***	−0.178***
	(0.015)	(0.015)	(0.018)	(0.013)	(0.012)	(0.015)
College	−0.033	−0.091**	−0.145***	−0.104***	−0.090***	−0.220***
	(0.028)	(0.038)	(0.036)	(0.022)	(0.019)	(0.026)
High School	−0.037**	0.014	−0.047**	−0.077***	−0.036***	−0.151***
	(0.016)	(0.017)	(0.019)	(0.013)	(0.012)	(0.015)
Minority	0.002	0.022	−0.020	−0.016	−0.024	0.003
	(0.029)	(0.041)	(0.046)	(0.035)	(0.030)	(0.036)
Prov FE	Y	Y	Y	Y	Y	Y
Occupation FE	Y	Y	Y	Y	Y	Y
Num. obs.	3741	4100	2973	3923	4066	4097

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Standard errors are robust and clustered at the township level. All models include provincial fixed effects.

Table A3. Correspondence Between Variable Keys and Survey Questions

Variable Key	Survey Question
Tax the Rich	Do you agree with the following statement? [More taxes should be imposed on the rich to help the poor]
Unemployment Priority	Do you think addressing [unemployment] should be given priority?
Decline in Socioeconomic Status	Compared with three years ago, has your socioeconomic status [declined]?
Life Worse	Compared with "the life of ordinary people," is your life [worse than the life of ordinary people]?
Workers Deserve High Pay	Among the following groups, do you think [workers] should receive higher incomes?
Not Employed Full Time	Main employment status within the past three months: [not employed full time].

Appendix E EIT Collection Jurisdiction by Ownership and Establishment Cohort

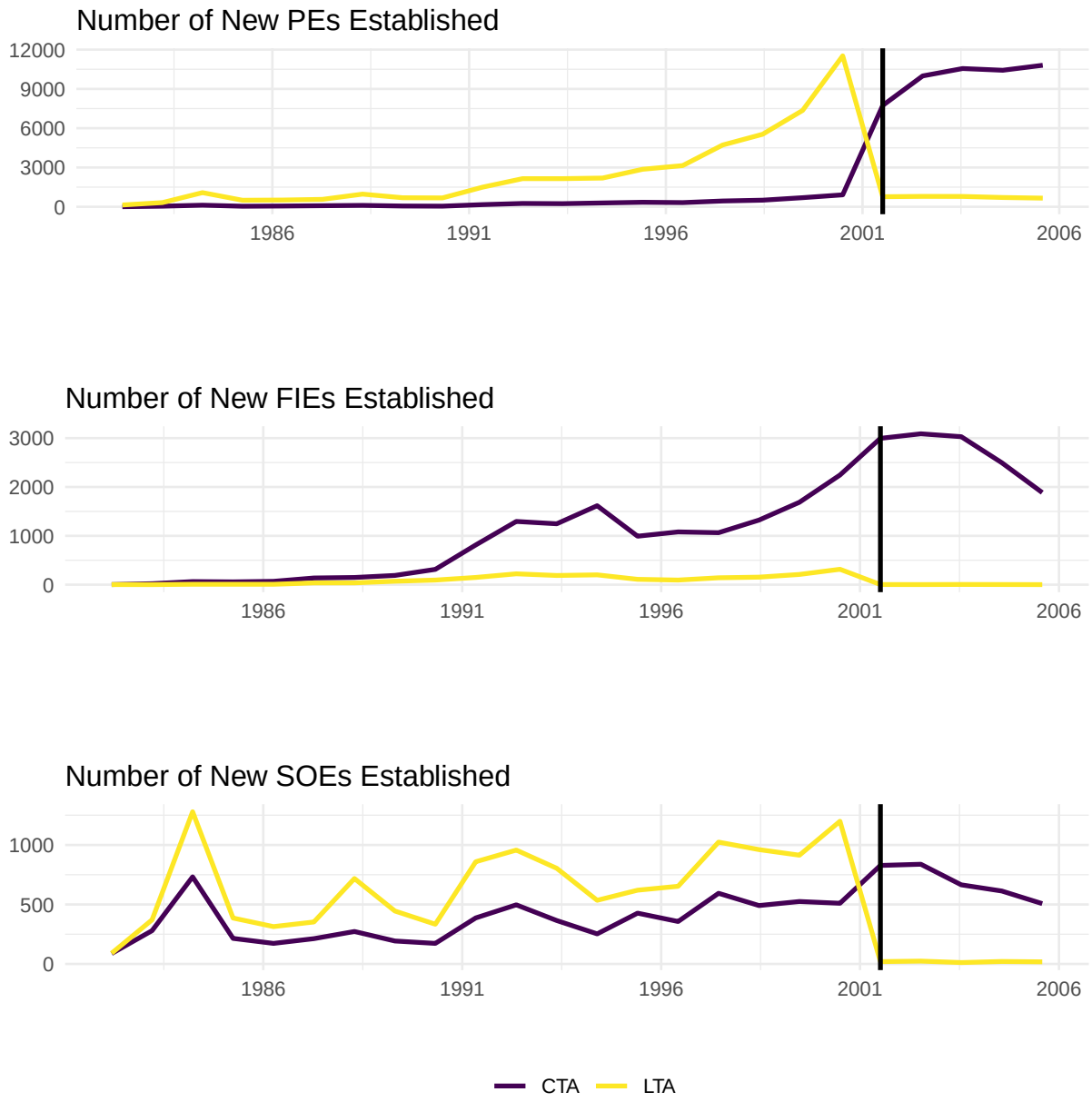


Figure A5. EIT collection jurisdiction by ownership type

Note. This figure uses the 2007 cross-section of the National Enterprise Tax Survey to report the tax agency responsible for EIT collection across ownership types. The y-axis represents the number of firms, and the x-axis groups firms by establishment year, 1986–2006. The vertical black line marks the 2002 EIT reform cutoff. Most new PEs and SOEs established in or after 2002 were administered by the CTA, while the plotted LTA cases show that the assignment was not categorical; FIEs were already principally administered by the CTA. Data source: National Enterprise Tax Survey, 2007.

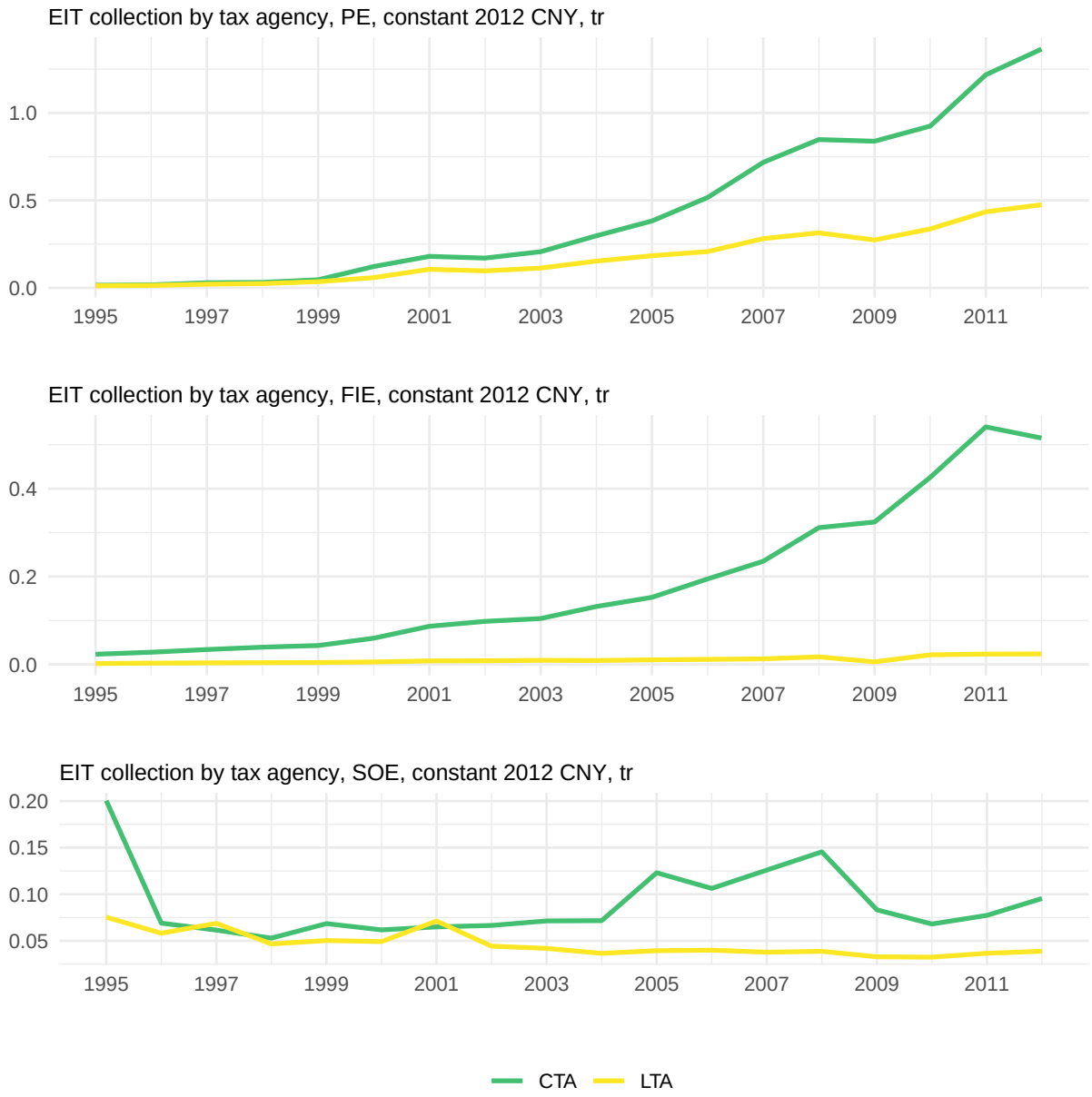


Figure A6. EIT collection by tax agency, 1995–2012

Note. The figure presents total EIT collections by the CTA and LTA across ownership types. The CTA began to collect a larger share of EIT from PEs and SOEs after 2002, consistent with the 2002 EIT reform. The CTA collects the EIT from nearly all FIEs. Data source: *China Taxation Yearbook*, various years.

Appendix F Constructing Tariff-Predicted Export Exposure

I construct the instrument for realized Bartik exposure in three steps. Following Bombardini and Li (2020), I estimate the relationship between destination-country tariffs and Chinese exports, recover fitted export levels, and aggregate their change to cities using pre-WTO output weights. Estimation uses product–destination variation rather than industry-average tariffs.

Let p index six-digit HS products, d destination markets, and t the initial or final period. Let X_{pdt} be the free-on-board (FOB) value of Chinese exports in product–destination cell pd, and let τ_{pdt} be the applied ad valorem tariff faced by that cell, measured in percentage points. The initial period pairs 2000 export values with the first available tariff observation, from 2001. The final period uses 2007 export values and tariffs. I estimate

$$\ln\left(X_{pdt}+1\right) = \theta \ln\left(1+\tau_{pdt}\right) + \eta_p + \lambda_d + \phi_t + \varepsilon_{pdt}, \tag{5}$$

where η_p, λ_d, and ϕ_t are product, destination, and period fixed effects. Zero export flows are retained through the one-unit offset, and standard errors are clustered by HS product. The regression contains 1,428,604 product–destination–period cells. The estimated tariff coefficient is $\widehat{\theta} = -0.0889$ (clustered standard error 0.0084). Because τ_{pdt} is recorded in percentage points, this coefficient is the elasticity of $X_{pdt}+1$ with respect to $1+\tau_{pdt}$ on that scale. It is not numerically comparable to a coefficient based on a tariff recorded as a decimal fraction.

Equation (5) makes explicit how tariff changes predict export increases. Holding the fixed effects constant, the tariff-induced fitted change for a product–destination cell is

$$\Delta\widehat{\ln(X+1)}_{pd,\tau} = \widehat{\theta}\left[\ln\left(1+\tau_{pd,2007}\right) - \ln\left(1+\tau_{pd,2001}\right)\right]. \tag{6}$$

Since $\widehat{\theta} < 0$, a tariff reduction makes the bracketed term negative and therefore contributes positively to predicted export growth. I then recover the fitted export-value level used in the construction:

$$\widehat{X}_{pdt} \equiv \exp\left[\widehat{\theta}\ln\left(1+\tau_{pdt}\right)+\widehat{\eta}_p+\widehat{\lambda}_d+\widehat{\phi}_t\right]. \tag{7}$$

This is the exponentiated fitted value of $\ln(X_{pdt}+1)$ used in the implementation. I apply the same back-transformation in both periods. As in Bombardini and Li (2020), the fitted level retains the estimated fixed effects. Thus "tariff-predicted" means predicted by the full tariff regression in Equation (5), not that the fitted level is a mechanical function of the tariff term alone.

I concord HS products through ISIC to the two-digit 2002 Chinese industry categories used for the city weights and sum fitted values over products and destinations:

$$\widehat{EXP}_{kt} = \sum_{p\in k}\sum_{d}\widehat{X}_{pdt}, \tag{8}$$

$$\widehat{\Delta EXP}_k^{00-07} = \widehat{EXP}_{k,2007} - \widehat{EXP}_{k,2000}. \tag{9}$$

Thus $\widehat{\Delta EXP}_k^{00-07}$ is a fitted change in export value. It is neither a tariff change nor an export growth rate, and its construction does not impose a unit elasticity between tariffs and exports.

Finally, I aggregate the fitted industry changes to cities using the same 2000 output weights and normalization as in realized export exposure:

$$\widetilde{EXPShock}_i^{07} = \sum_k \frac{Y_{ik}^{00}}{Y_k^{00}}\frac{\widehat{\Delta EXP}_k^{00-07}}{Y_i^{00}} \tag{10}$$

$$= \sum_k \frac{Y_{ik}^{00}}{Y_i^{00}}\frac{\widehat{\Delta EXP}_k^{00-07}}{Y_k^{00}}. \tag{11}$$

The second line displays the measure in the shift–share form used by Bombardini and Li (2020): a city's initial output share in industry k multiplies the national tariff-predicted export change per unit of that industry's initial output. In the source data, export values are in US dollars and 2000 gross industrial output from the Annual Survey of Manufacturing Firms is in thousands of yuan. The

instrument is therefore measured as fitted export dollars per thousand yuan of baseline industrial output. The realized and predicted measures differ only in whether ΔEXP_k^{00-07} or $\widehat{\Delta EXP}_k^{00-07}$ enters the common aggregation, so their units and normalization coincide.

The exclusion restriction applies to the fitted-export instrument as implemented. Conditional on baseline city characteristics, regional fixed effects, and baseline industrial-structure controls, cross-city differences in fitted export exposure from Equation (11) must affect the outcomes through realized export exposure rather than through an independent city-specific path. Because the fitted export levels retain the estimated fixed effects, this assumption concerns the full fitted-export measure, not the tariff term in isolation. The mechanism tests then ask where that exposure response appears: new private firms, CTA recruitment in local branch offices, and tax-specific fiscal-book revenue. The fiscal contrasts use categories aligned with the legal collection and sharing rules. The center's EIT and VAT shares are calculated from fiscal-book revenue using those rules.

Appendix G Measuring Profit-Reporting Alignment

This section examines profit reporting among firms presumptively assigned to the CTA under the 2002 default rule. I compare reported profits with an imputed benchmark and examine how their alignment changed from 2002 to 2007. This exercise complements the city analysis by asking whether reporting in the presumptive CTA sample moved in the direction expected under stronger tax administration.

Following H. Cai and Liu (2009), I construct a profit benchmark by subtracting intermediate inputs, financing costs, wages, current depreciation, and VAT from gross output, then compare that benchmark with reported profit. The procedure adapts book–tax-difference approaches used to study corporate reporting and avoidance (M. A. Desai & Dharmapala, 2006; Manzon & Plesko, 2002). Because the benchmark is constructed rather than observed, the resulting gap measures reporting alignment, not tax evasion directly.

Let $PRO_{i,t}$ represent the imputed profit of firm i in year t, $RPRO_{i,t}$ represent the reported profit, and $\pi_{i,t}$ denote the real (unobserved) profit. The imputed profit is calculated as

$$PRO_{i,t} = Y_{i,t} - MED_{i,t} - FC_{i,t} - WAGE_{i,t} - CURRD_{i,t} - VAT_{i,t},$$

where the variables are defined as follows:

- $Y_{i,t}$ is the firm's gross output.
- $MED_{i,t}$ measures intermediate inputs excluding financial costs.
- $FC_{i,t}$ represents financing costs.
- $WAGE_{i,t}$ denotes total wages.
- $CURRD_{i,t}$ is current depreciation.
- $VAT_{i,t}$ is the firm's VAT payment.

The imputed profit $PRO_{i,t}$ is a noisy estimate of real profit $\pi_{i,t}$ and is expected to be positively correlated with it. The relationship is modeled as

$$\pi_{i,t} = PRO_{i,t} + \eta_{i,t} + \theta_{i,t},$$

where $\eta_{i,t}$ captures differences in accounting standards across firms[1] and $\theta_{i,t}$ is a random error with mean zero.

The relationship between reported and real profits is then given by

$$RPRO_{i,t} = d_{i,t}\pi_{i,t} + e_{i,t} + \xi_{i,t},$$

[1]For instance, firms may follow different depreciation rates, leading to discrepancies between reported and real profits.

where $d_{i,t} < 1$ measures the alignment between reported and real profit. A smaller $d_{i,t}$ indicates greater misalignment. $e_{i,t} < 0$ is consistent with underreporting, and $\xi_{i,t}$ is a random error term with mean zero.

Substituting $\pi_{i,t}$ into the equation for reported profit yields

$$RPRO_{i,t} = d_{i,t} PRO_{i,t} + d_{i,t}\eta_{i,t} + e_{i,t} + \varepsilon_{i,t},$$

where $\varepsilon_{i,t}$ is a random error term. The coefficient $d_{i,t}$ represents reporting alignment: a higher $d_{i,t}$ implies a smaller difference between benchmark and reported profit.

G.1 Data and Econometric Specification

I use an unbalanced panel of China-based manufacturing firms from the ASMF, administered by China's National Bureau of Statistics, for 2002 to 2007. The survey covers all SOE manufacturers and non-SOE manufacturers with annual sales above CNY 5 million. Its financial variables permit comparison of reported and imputed profits over time.

I follow *Generally Accepted Accounting Principles* (GAAP) screening procedures to remove erroneous observations.[2] The ASMF does not identify the collecting agency, and these records are not merged with a separate jurisdiction source. I infer CTA jurisdiction from establishment cohort, retaining firms established on or after January 1, 2002. Their EIT generally fell under CTA collection by default. This criterion cannot identify mergers, divisions, or ownership conversions that remained with the LTA under statutory exceptions. The estimates therefore describe reporting alignment in a rule-inferred CTA sample.[3]

[2] I remove records with missing key variables, fewer than ten employees, accounting errors such as fixed assets exceeding total assets, and duplicate observations.

[3] Appendix Table A1 summarizes the default collection rule. Appendix Figure A5 uses the separate 2007 National Enterprise Tax Survey, which observes collecting agencies, to show the CTA and LTA distribution by ownership and establishment cohort.

Table A4. Profit-Reporting Alignment, 2002–2007

	2002	2003	2004	2005	2006	2007
Imputed profit	0.075***	0.096***	0.099***	0.111***	0.113***	0.120***
	(0.012)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)
City FE	Y	Y	Y	Y	Y	Y
Ind FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y
Num. obs.	1376	9145	23131	30812	65196	88469

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Standard errors are robust and clustered at the city level. Control variables include firm age and total current assets.

G.2 Results

To estimate how $d_{i,t}$ varies over time from 2002 to 2007, I run the following specification in the presumptive CTA sample:

$$RPRO_{i,m,n} = \gamma_1 PRO_{i,m,n} + \gamma_2 X_{i,m,n} + \omega_m + \xi_n + \varepsilon_{i,m,n},$$

where i, m, and n index firms, cities, and industries, respectively. $X_{i,m,n}$ includes firm-level covariates such as firm age and size. I include industry and city fixed effects, and $\varepsilon_{i,m,n}$ is the error term.

The key coefficient of interest is γ_1, which measures $d_{i,t}$. A higher γ_1 implies closer alignment between benchmark and reported profit. A lower value implies a wider reporting gap.

I report the estimates of γ_1 for 2002–2007 in Table A4. The coefficient rises by roughly 60% across the period, indicating that reported profits moved closer to the imputed benchmark in this sample. The pattern is consistent with improved reporting alignment among firms presumptively assigned to the CTA and provides a firm-level counterpart to the city results. As a national repeated-cross-section pattern based on rule-inferred jurisdiction, it corroborates the proposed mechanism without serving as a separate estimate of the effect of local CTA staffing.



Figure A7. Profit-reporting alignment, presumptive CTA sample, 2002–2007

Note. This figure presents coefficient estimates from regressions of reported profit on the constructed profit benchmark. A higher positive coefficient indicates closer reporting alignment. Coefficients are estimated using the full specifications.

I interpret the upward trend in Figure A7 as consistent with improving profit reporting among firms presumptively assigned to the CTA. The gap between imputed and reported profits narrowed in the observed sample during the same institutional period in which the CTA expanded.

Appendix H Additional Identification and Robustness Analyses

The supplementary analyses examine inference, exposure construction, revenue measurement, outcome scale, timing, fiscal compensation, and influential observations. Each set of checks addresses a specific concern about the entry, recruitment, revenue, or fiscal compensation results. Tables A17–A19 use HC2 standard errors for transformation, sector-composition, and pre-WTO placebo diagnostics with one observation per city. These checks complement the province-clustered main inference.

H.1 Data Summary and Inference Diagnostics

Table A5 summarizes the variables underlying the main tables. Table A6 re-estimates the headline Bartik specifications with city-clustered standard errors. Its EIT column retains the outcome, controls, and 310-city sample of Table 3 in the main text. Only the standard-error calculation changes. With one observation per city, city clustering is equivalent to robust inference. Point estimates remain unchanged, and these standard errors allow no province-level dependence.

Table A7 bootstraps the first stage and several OLS associations with realized exposure at the province level. Because realized exposure is endogenous, the outcome rows are neither reduced forms nor wild-bootstrap IV tests. They assess directional consistency under a different small-cluster calculation and complement the province-clustered 2SLS inference.

Table A5. Summary Statistics for Main Empirical Variables

Variable	N	Missing	Mean	SD	P25	Median	P75	Max
Realized export exposure (firm sample)	310	2	112.423	55.598	74.757	111.557	152.408	253.703
Tariff-predicted exposure (firm sample)	310	2	36.321	26.086	13.460	32.937	52.614	109.136
Realized export exposure (CTA recruitment sample)	312	0	112.051	55.783	74.386	111.131	151.938	253.703
Δ Private firms	310	2	995.248	1731.536	237.000	492.000	888.000	16659.000
Δ Private-sector employment	251	61	115.921	219.406	11.300	50.684	154.945	1693.329
Δ FIE firms	310	2	31.284	134.720	0.000	7.500	20.750	2107.000
Δ SOE firms	310	2	-25.165	62.612	-44.000	-19.500	2.000	630.000
All CTA positions	312	0	7.968	14.236	0.000	3.000	12.000	130.000
Mean CTA positions, 2006–2008	312	0	5.159	7.470	1.500	3.500	5.750	77.500
Branch CTA positions	312	0	7.638	13.674	0.000	3.000	11.250	126.000
Headquarters CTA positions	312	0	0.330	3.904	0.000	0.000	0.000	65.000
Log EIT share	310	2	9.833	1.706	8.874	9.809	10.892	14.817
Log VAT share	310	2	11.563	1.349	10.865	11.611	12.366	15.260
Log [EIT / (VAT + consumption tax)]	309	3	-2.099	0.732	-2.575	-2.123	-1.578	-0.091
EIT / (EIT + VAT) (pp)	310	2	28.080	11.966	18.957	27.299	36.432	70.911

Summary statistics use the included analysis-ready city sample. Export exposures are measured in US dollars per thousand yuan of baseline industrial output. The first two rows use the 310-city private-firm sample; the realized-exposure standard deviation in that sample is 55.598, rounded to 55.6 in the private-firm discussion. The third row uses the 312-city CTA recruitment sample; its realized-exposure standard deviation is 55.783 and its 74.386–151.938 interquartile range is the 74.4–151.9 spread used to interpret the CTA-recruitment estimates. Firm variables are 2000–2007 changes; CTA outcomes summarize 2007 recruitment, except the stated 2006–2008 mean. The center's EIT share equals 1.5 times fiscal-book EIT, and its VAT share equals three times fiscal-book VAT. The logged EIT / (VAT + consumption tax) ratio is calculated by dividing the center's EIT share by the sum of the reported 75% VAT remittance and consumption-tax revenue and then taking the log. These revenue amounts are calculated from fiscal books using the legal sharing rules, not observed agency collections. Fiscal amount outcomes are logged 2007 values; EIT / (EIT + VAT) is measured in percentage points.

Table A6. City-Clustered Standard Errors

	Firm Registration	CTA Recruitment	Tax Revenue
	Δ PE	CTA positions	EIT
Export Exposure	8.247***	0.075***	0.005**
	(2.585)	(0.029)	(0.002)
Region FE	Y	Y	Y
Controls	Y	Y	Y
Cluster	City	City	City
Num. obs.	310	312	310

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: the private-firm outcome is a count change; CTA recruitment is measured in position counts; EIT uses source fiscal-statistics currency units before logging. Transformations: EIT is logged; export exposure is instrumented with tariff-predicted export exposure; no outcome is converted to per-capita units. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects in every column. SE: standard errors clustered by city as a diagnostic with one observation per city. Sample: included city-level cross-section with one observation per city. Years: firm changes are measured from 2000 to 2007; fiscal outcomes are measured in 2007; controls are measured in 2000. Outcome scale: private-firm changes and CTA recruitment positions are levels; EIT is a log outcome; no outcome is IHS-transformed or expressed per capita. Each city appears once, so city-clustered inference is equivalent to robust inference in this cross-sectional setting. The EIT column re-estimates the headline EIT specification on the same 310-city sample; only the standard-error calculation changes.

Table A7. Provincial Wild-Cluster Bootstrap OLS Checks for Bartik Outcomes

Model	Term	Estimate	Wild-bootstrap SE	p-value	N
First stage	`exp_shock_pred_p`	1.679	0.105	< 0.001	312
Firm registration OLS check	`exp_shock_raw_p`	5.211	1.530	< 0.001	310
CTA recruitment OLS check	`exp_shock_raw_p`	0.060	0.025	0.015	312
EIT OLS check	`exp_shock_raw_p`	0.004	0.002	0.029	310

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. The table reports provincial wild-cluster bootstrap checks using 500 bootstrap replications and clusters defined by `prov_cd`. The first row bootstraps the first stage. The outcome rows regress each outcome on realized export exposure with regional fixed effects and baseline controls. Because realized exposure is endogenous, those rows are OLS association checks rather than reduced forms or bootstrapped IV estimates. They are supplemental diagnostics, not replacements for the province-clustered 2SLS estimates.

H.2 Identification and Exposure Checks

In shift-share designs, a small set of industries can account for much of the identifying variation. Goldsmith-Pinkham et al. (2020) propose Rotemberg weights to measure each initial industry share's influence on the estimate. Higher weights identify more influential industries.

Following their recommendation, I compute Rotemberg weights for export exposure. Table A8 presents the Rotemberg weights of the top 10 industries for export exposure. To assess whether the most influential industry shares mechanically drive the results, I reconstruct export exposure and its tariff-predicted instrument after excluding the five industries with the highest Rotemberg weights. Table A9 keeps the original exposure measure as the comparison column, then rebuilds the robustness exposure measure using baseline industry exposure data and removes the top industries cumulatively. The private-firm and CTA-recruitment estimates remain near 8.7 and 0.078, respectively. The estimates for the center's EIT share remain positive, with coefficients near 0.004 after the first exclusion, and those for its VAT share remain close to zero. The broader bundle of EIT and individual income tax (IIT) remains positive and statistically significant at the 10% level after every exclusion, with coefficients near 0.004. Tables A10, A11, and A12 provide additional no-top-Rotemberg results. The stable signs across the sequential exclusions show that no single high-weight industry generates the institutional sequence.

Table A8. Top 10 Industries: Rotemberg Weights

Industry code	Industry	β	α_k
356	Furnace and electric furnace manufacturing	0.2066	0.3583
359	Metal casting and forging	0.0696	0.2328
282	Synthetic fiber manufacturing	0.2425	0.2023
137	Fruit, vegetable, and nut processing	0.2693	0.1805
233	Reproduction of recorded media	0.3208	0.1656
376	Aircraft and spacecraft manufacturing	0.0881	0.1630
133	Vegetable oil processing	0.1649	0.1426
266	Special chemical products manufacturing	0.0218	0.1424
171	Cotton and chemical fiber textile processing	0.1002	0.1334
371	Railway transport equipment manufacturing	0.2391	0.1318

Units: β is the industry-specific Rotemberg coefficient and α_k is the Rotemberg weight share. Industry labels are concise English translations of the GB/T 4754–2002 industry names. Transformations: weights are computed from residualized city-level exposure and fiscal-capacity variation using baseline firm-output shares; industry shifts are recovered from the saved main city-level shock so the full reconstructed exposure equals the paper's main export exposure; no outcome is logged, IHS-transformed, or expressed per capita in this descriptive table. Controls: residualization uses baseline 2000 unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects enter the residualization step. SE: no regression standard errors are reported because this is a descriptive weight table. Sample: included city-level cross-section matched to upstream industry firm-output shares. Years: baseline shares and controls are measured in 2000; the saved main export exposure measures the paper's 2000–2007 exposure window. Outcome scale: descriptive weights and coefficients are in levels; no table outcome is IHS-transformed or expressed per capita.

Table A9. Sequential Rotemberg Exclusions: 2SLS Estimates

	Full main exposure	Drop top 1 +356 Furnaces	Drop top 1–2 +359 Casting/forging	Drop top 1–3 +282 Synthetic fibers	Drop top 1–4 +137 Fruit/veg/nuts	Drop top 1–5 +233 Recorded media
Δ Private firms	8.247***	8.675***	8.673***	8.668***	8.668***	8.668***
	(2.331)	(2.402)	(2.401)	(2.401)	(2.401)	(2.401)
CTA positions, 2007	0.075***	0.077***	0.077***	0.078***	0.078***	0.078***
	(0.027)	(0.028)	(0.028)	(0.028)	(0.028)	(0.028)
EIT share	0.005**	0.004	0.004	0.004	0.004	0.004
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
VAT share	0.002	0.001	0.001	0.001	0.001	0.001
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
EIT + IIT (appendix)	0.005**	0.004**	0.004**	0.004**	0.004**	0.004**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Region FE	Y	Y	Y	Y	Y	Y
PCA Ind	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: export exposures are measured in US dollars per thousand yuan of baseline industrial output; private firms are count changes; CTA positions in 2007 are counts; the center's EIT share equals 1.5 times fiscal-book EIT, the center's VAT share equals three times fiscal-book VAT, and the EIT-plus-IIT row reports a sensitivity measure equal to 1.5 times the sum of fiscal-book EIT and individual income tax (IIT); these revenue amounts are calculated from the legal sharing rules, not observed agency collections. Transformations: the center's EIT and VAT shares and the appendix EIT-plus-IIT bundle are logged; the first column uses the original main export exposure and instrument, while each exclusion column instruments its reconstructed export exposure with the corresponding tariff-predicted exposure; top-weighted industries are cumulatively excluded one at a time. Controls: full baseline 2000 census and industry controls in every column. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section with cumulative top-Rotemberg industry exclusions. Years: firm changes are measured from 2000 to 2007; fiscal outcomes are measured in 2007; controls are measured in 2000. Outcome scale: private firms and CTA positions are level outcomes; the center's EIT and VAT shares and the appendix EIT-plus-IIT bundle are log outcomes; no outcome is IHS-transformed or expressed per capita. Each cell reports a separate full-control 2SLS coefficient for the outcome in that row. The first column uses the original main export exposure for comparison; subsequent columns cumulatively exclude the top Rotemberg-weight industries listed in Table A8 from the robustness shock, one additional industry at a time. Sequential exclusion order: top 1 removes 356 (Furnace and electric furnace manufacturing); top 2 adds 359 (Metal casting and forging); top 3 adds 282 (Synthetic fiber manufacturing); top 4 adds 137 (Fruit, vegetable, and nut processing); top 5 adds 233 (Reproduction of recorded media).

Table A10. Export Exposure and Change in Firm Registration: 2SLS Estimates with Top Five Rotemberg Industries Removed

	First Stage	Δ PE			Δ PE Emp	Δ FIE	Δ SOE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Export Exposure		10.400***	11.696***	8.242***	1.149**	0.275**	0.038
		(2.812)	(2.954)	(2.332)	(0.419)	(0.130)	(0.104)
$\widetilde{EXPShock_i}^{07}$	1.679***						
	(0.105)						
Region FE	Y	Y	Y	Y	Y	Y	Y
PCA Ind	N	N	Y	Y	Y	Y	Y
Controls	N	N	N	Y	Y	Y	Y
Num. obs.	312	310	310	310	251	310	310

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: export exposures are measured in US dollars per thousand yuan of baseline industrial output; firm outcomes are counts, except that private-sector employment is measured in thousands. Transformations: the no-top export exposure is instrumented with the no-top tariff-predicted exposure; private-sector employment is divided by 1,000; no firm outcome is logged in this robustness table. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure; columns marked PCA include only the principal components. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section with top Rotemberg industries removed from the shock. Years: firm changes are measured from 2000 to 2007; controls are measured in 2000. Outcome scale: outcomes are level changes; no outcome is IHS-transformed or expressed per capita. Because these city-level models are cross-sectional with one observation per city, city-level clustering is not separately informative; province-level clustering allows arbitrary correlation in unobserved shocks across cities within the same province. The export exposure and instrument exclude the five highest-Rotemberg-weight industries listed in Table A8; the no-top shocks are reconstructed from baseline firm-output shares by subtracting each excluded industry's contribution from the robustness shock.

Table A11. Export Exposure and CTA Recruitment Positions: 2SLS Estimates with Top Five Rotemberg Industries Removed

	All positions			06–08 Mean	Branch	Headquarters
	(1)	(2)	(3)	(4)	(5)	(6)
Export Exposure	0.107***	0.099***	0.076***	0.038**	0.077***	−0.001
	(0.027)	(0.031)	(0.027)	(0.015)	(0.027)	(0.002)
Region FE	Y	Y	Y	Y	Y	Y
PCA Ind	N	Y	Y	Y	Y	Y
Controls	N	N	Y	Y	Y	Y
Num. obs.	312	312	312	312	312	312

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: tax-administration outcomes are CTA recruitment-position counts; column 4 is the 2006–2008 mean count; columns 5–6 split branch and headquarters counts. Transformations: outcomes are unlogged levels; the no-top export exposure is instrumented with the no-top tariff-predicted exposure. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure; columns marked PCA include only the principal components. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section with top Rotemberg industries removed from the shock. Years: main CTA recruitment outcomes are measured in 2007, except the mean column uses 2006–2008; controls are measured in 2000. Outcome scale: all outcomes are levels; no outcome is IHS-transformed or expressed per capita. Because these city-level models are cross-sectional with one observation per city, city-level clustering is not separately informative; province-level clustering allows arbitrary correlation in unobserved shocks across cities within the same province. The export exposure and instrument exclude the five highest-Rotemberg-weight industries listed in Table A8; the no-top shocks are reconstructed from baseline firm-output shares by subtracting each excluded industry's contribution from the robustness shock.

Table A12. Export Exposure, Firm Entry, and Central Revenue with Top Five Rotemberg Industries Removed

	Dependent variable: Log EIT share				Dependent variable: Log VAT share			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Export Exposure	0.005**				0.002			
	(0.002)				(0.002)			
Δ PE		0.319***				0.212***		
		(0.079)				(0.054)		
Δ FIE			1.373				0.871	
			(0.837)				(0.564)	
Δ SOE				−1.490				−1.841**
				(1.049)				(0.780)
Region FE	Y	Y	Y	Y	Y	Y	Y	Y
PCA Ind	Y	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Num. obs.	310	308	308	308	310	308	308	308

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: the center's EIT share equals 1.5 times fiscal-book EIT under the 60:40 sharing rule, and the center's VAT share equals three times fiscal-book VAT under the 75:25 sharing rule; these revenue amounts are calculated from the legal sharing rules, not observed agency collections; firm-sector regressors are firm-count changes in thousands. Transformations: the center's EIT and VAT shares are logged; firm-sector regressors are calculated by dividing firm-count changes by 1,000; columns 1 and 5 instrument the no-top export exposure with the no-top tariff-predicted exposure, while the remaining columns report controlled OLS associations. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure; columns marked PCA include only the principal components. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section with top Rotemberg industries removed from the shock. Years: fiscal outcomes are measured in 2007; firm changes are measured from 2000 to 2007; controls are measured in 2000. Outcome scale: all outcomes are log levels; no outcome is IHS-transformed or expressed per capita. Because these city-level models are cross-sectional with one observation per city, city-level clustering is not separately informative; province-level clustering allows arbitrary correlation in unobserved shocks across cities within the same province. The export exposure and instrument exclude the five highest-Rotemberg-weight industries listed in Table A8; the no-top shocks are reconstructed from baseline firm-output shares by subtracting each excluded industry's contribution from the robustness shock.

Table A13 then varies the exposure measure and estimating sample for private-firm entry, CTA recruitment, and EIT's share of EIT plus VAT. Column 1 reproduces the main 2SLS specification. Column 2 uses the exposure measure through 2010, while column 3 enters tariff-predicted exposure directly as a reduced-form regressor. Column 4 reconstructs both realized export exposure and its tariff-predicted instrument from the underlying industry-level components and cities' 2000 output weights, using the same sample, controls, and regional fixed effects. This reconstruction provides the baseline for column 5, which removes the industry with the highest Rotemberg weight. Columns 6 and 7 exclude coastal province-level units and provincial capitals and municipalities, respectively. Column 8 replaces region fixed effects with province fixed effects. The three outcomes retain the predicted positive sign in every variant. The firm and recruitment estimates are especially stable. The EIT–VAT composition estimate is positive and statistically significant at the 5% level in seven specifications. It is significant at the 10% level when provincial capitals and municipalities are excluded. This pattern is consistent with a shift toward the tax base directly affected by the 2002 reform.

Table A13. Alternative Exposure Specifications

	Main 2SLS	2010 exposure	Predicted exposure	Industry shock	Drop top ind.	No coastal	No capital/muni.	Province FE
Δ Private firms	8.247***	4.949***	13.395***	8.675***	8.675***	5.339**	2.032**	9.389***
	(2.331)	(1.508)	(3.669)	(2.402)	(2.402)	(2.507)	(0.894)	(2.021)
CTA positions	0.075***	0.048***	0.123***	0.077***	0.077***	0.068**	0.051***	0.034
	(0.027)	(0.017)	(0.044)	(0.028)	(0.028)	(0.032)	(0.016)	(0.025)
EIT/(EIT+VAT) share (pp)	0.059***	0.039***	0.099***	0.060***	0.060***	0.055**	0.033*	0.082***
	(0.017)	(0.012)	(0.029)	(0.018)	(0.018)	(0.024)	(0.018)	(0.018)
Estimator	2SLS	2SLS	OLS	2SLS	2SLS	2SLS	2SLS	2SLS
Region FE	Y	Y	Y	Y	Y	Y	Y	N
Province FE	N	N	N	N	N	N	N	Y
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Sample	All	All	All	All	All	No coast	No capital/muni.	All

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: export exposures are measured in US dollars per thousand yuan of baseline industrial output; private firms and CTA recruitment are count outcomes; EIT/(EIT+VAT) is measured in percentage points. Transformations: columns 1, 2, 4, 5, 6, 7, and 8 instrument raw exposure with the corresponding tariff-predicted exposure; column 3 enters tariff-predicted exposure directly; column 2 uses the ready-data 2010 exposure variant; columns 4 and 5 use reconstructed industry shocks; column 5 excludes the highest-Rotemberg-weight industry from the shock. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects in columns 1–7 and province fixed effects in column 8. SE: standard errors clustered by province. Sample: included city-level cross-section; column 6 excludes coastal province-level units; column 7 excludes provincial capitals and municipalities. Years: firm changes are measured from 2000 to 2007; fiscal outcomes are measured in 2007; controls are measured in 2000. Outcome scale: private firms and CTA recruitment are level outcomes; EIT/(EIT+VAT) is a percentage-point share outcome; no outcome is IHS-transformed or expressed per capita. Additional check for alternative exposure and sample specifications. Coastal province-level units are province codes 12, 13, 21, 31, 32, 33, 35, 37, 44, 45, and 46. Provincial capitals are identified by city-code suffix 01; municipalities are province codes 11, 12, 31, and 50.

H.3 Revenue Measurement and Fiscal-Composition Checks

Table 3 in the main text reports the center's EIT and VAT shares and two comparisons of EIT with other tax revenue. Appendix Table A14 shows that business and construction taxes also grew with exposure. This is expected as exporting expands the broader tax base. The institutional test concerns composition: EIT rises relative to VAT, whose collection rules were unchanged in 2002. An exclusive increase in EIT is not required.

Appendix Table A15 tests the center's logged EIT and VAT shares under alternative outlier treatments. Appendix Table A16 examines baseline and pre-trend controls, winsorization, and log-odds and rank transformations. The legal sharing rules determine the center's EIT and VAT shares from fiscal-book revenue. Collection rules place most new firms under the CTA. The broader EIT-plus-IIT bundle provides an additional income-tax sensitivity check.

Table A14. Export Exposure and the Local-Tax Benchmark

	Business Tax + Construction Tax
Export Exposure	0.005***
	(0.002)
Effect across exposure IQR	+52.9%
Num. obs.	310
First-stage F	139.4
Region FE	Y
PCA industry controls	Y
Baseline census controls	Y

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. The table reports a city-level 2SLS regression of the logged 2007 local-tax benchmark on realized export exposure, instrumented by tariff-predicted exposure. The benchmark is the sum of fiscal-book business tax and construction tax, categories principally administered by the LTA during the sample period. The model includes regional fixed effects, the first three principal components of baseline industrial structure, and baseline census controls; standard errors are clustered by province. The interquartile-range row converts the logged effect to a percent change. City fiscal books identify tax categories rather than agency-coded receipts, so this is an assignment-based benchmark, not observed total LTA collections.

Table A15. City-Level Central Revenue and Outlier Diagnostics

	Full log	Log p1/p99	Log p5/p95
EIT share	0.005**	0.005**	0.004*
	(0.002)	(0.002)	(0.002)
VAT share	0.002	0.002	0.001
	(0.002)	(0.002)	(0.002)
EIT sample size	310	310	310
VAT sample size	310	310	310
Region FE	Y	Y	Y
Province-clustered SE	Y	Y	Y

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Each cell reports a separate city-level 2SLS estimate of realized export exposure, instrumented by tariff-predicted exposure. All models include regional fixed effects and baseline census and industrial controls, with standard errors clustered by province. The center's EIT share equals 1.5 times fiscal-book EIT under the 60:40 sharing rule, and its VAT share equals three times fiscal-book VAT under the 75:25 sharing rule. These revenue amounts are calculated from the legal sharing rules, not observed agency collections. The columns use logged amounts. Winsorization is applied once to raw 2007 amounts at the stated cutoffs.

Table A16. Export Exposure and EIT–VAT Tax Composition

	Log EIT share	EIT / (EIT + VAT)	+ Baseline share	Winsorized share
Panel A. Amount, share, and baseline specifications				
Export Exposure	0.005**	0.059***	0.057***	0.060***
	(0.002)	(0.017)	(0.016)	(0.017)
Effect across exposure IQR	+48.4%	+4.64 pp	+4.46 pp	+4.67 pp
Region FE	Y	Y	Y	Y
PCA industry controls	Y	Y	Y	Y
Baseline census controls	Y	Y	Y	Y
Num. obs.	310	310	309	310
First-stage *F*	139.4	139.4	138.9	139.4

	Log EIT/VAT	+ Baseline ratio	+ Baseline and pre-trend	Share log-odds	Share rank
Panel B. Ratio and transformation specifications					
Export Exposure	0.003***	0.003***	0.002**	0.003***	0.005***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Effect across exposure IQR	+28.1%	+22.3%	+20.8%	+0.25 log-odds	+0.39 SD
Region FE	Y	Y	Y	Y	Y
PCA industry controls	Y	Y	Y	Y	Y
Baseline census controls	Y	Y	Y	Y	Y
Num. obs.	310	301	264	310	310
First-stage *F*	139.4	140.5	104.2	139.4	139.4

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Each cell reports a city-level 2SLS coefficient on realized export exposure, instrumented by tariff-predicted exposure. Standard errors are clustered by province. Baseline census controls are unemployment share, share of the population employed in manufacturing, the share of the population aged 20–34, and the male share; industry controls are the first three principal components of baseline industrial structure. Panel A reports the log of the center's EIT share, calculated as 1.5 times fiscal-book EIT, the 2007 EIT share of EIT plus VAT, the same share controlling for its 2000 value, and the share winsorized at the 1st and 99th percentiles. Panel B reports the log EIT-to-VAT ratio, specifications that add its 2000 value and its 1997–2000 pre-trend, and log-odds and inverse-normal-rank transformations of EIT's share of combined EIT and VAT revenue. The IQR row uses each column's estimation-sample exposure range; logged outcomes are converted to percent changes, share outcomes to percentage points, and the transformed-share outcomes to their indicated scales. Fiscal-book categories do not include intergovernmental transfers or identify the collecting agency. The 2002 reform brought EIT liabilities for most newly established firms under CTA administration and central revenue sharing; VAT was already centrally administered, and its assignment did not change in that reform.

H.4 Transformations, Per-Capita Scaling, and Sector Composition

I examine sensitivity to outcome scale, skewed count changes, and ownership composition. Table A17 reports log, IHS, and per-capita versions of the firm-entry and CTA-recruitment outcomes. Table A18 enters PE, FIE, and SOE firm changes jointly to examine whether the EIT association is specific to private-firm entry or reflects broader firm growth.

Table A17. Outcome Transformations and Per-Capita Scaling

	Firm Registration			CTA Recruitment
	Log PE	IHS PE	PE Per Cap.	Log CTA positions
Export Exposure	0.007***	0.007***	0.008***	0.006**
	(0.002)	(0.002)	(0.003)	(0.002)
Region FE	Y	Y	Y	Y
Controls	Y	Y	Y	Y
SE	HC2	HC2	HC2	HC2
Num. obs.	310	310	310	312

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: firm-registration outcomes are count changes; the CTA recruitment outcome is measured in position counts; per-capita private-firm registration is per 10,000 baseline residents. Transformations: log outcomes are log(y + 1) for nonnegative changes, with negative changes set to missing only for log diagnostics; IHS denotes inverse hyperbolic sine and preserves negative changes; per-capita private-firm registration is calculated by dividing private-firm changes by baseline population and multiplying by 10,000. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects in every column. SE: HC2 heteroskedasticity-robust standard errors. Sample: included city-level cross-section with one observation per city. Years: firm changes are measured from 2000 to 2007; the CTA recruitment outcome is measured in 2007; controls are measured in 2000. Outcome scale: columns 1 and 4 are log outcomes; column 2 is IHS; column 3 is expressed per capita. Additional check of outcome transformations and per-capita scaling.

Table A18. Firm-Sector Composition and EIT Revenue

	Log EIT
PE	0.468***
	(0.074)
FIE	−2.422***
	(0.782)
SOE	−0.263
	(1.302)
Region FE	Y
Controls	Y
SE	HC2
Num. obs.	308

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: EIT uses source fiscal-statistics currency units before logging; PE, FIE, and SOE regressors are firm-count changes in thousands. Transformations: EIT is logged; PE, FIE, and SOE changes are divided by 1,000 and entered jointly. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects in every column. SE: HC2 heteroskedasticity-robust standard errors. Sample: included city-level cross-section with one observation per city. Years: EIT is measured in 2007; firm changes are measured from 2000 to 2007; controls are measured in 2000. Outcome scale: EIT is a log-level outcome; no outcome is IHS-transformed or expressed per capita. Additional check of firm-sector composition and EIT revenue.

H.5 Placebo and Pre-Trend Tests

In Table A19, I use changes from 1997 to 2000 rather than baseline levels to test whether future predicted exposure is already associated with pre-treatment fiscal growth. Most placebo estimates are small and statistically insignificant, although the EIT placebo is positive at the 10% level. Table A20 reports the restructuring-side leads. The estimated association between later SOE employment-share decline and earlier grant growth is positive but imprecise ($p = 0.103$). The social-spending lead is near zero.

Table A19. Pre-WTO Placebo Tests

	Pre-WTO Change, 1997–2000		
	Revenue	EIT	Transfers
Tariff-predicted exposure	0.000	0.005*	0.001
	(0.001)	(0.003)	(0.001)
Region FE	Y	Y	Y
Controls	Y	Y	Y
SE	HC2	HC2	HC2
Num. obs.	305	264	305

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: revenue, EIT, and transfer variables use source fiscal-statistics currency units before logging. Transformations: outcomes are pre-WTO changes measured as log(y2000 + 1) minus log(y1997 + 1); the regressor is tariff-predicted export exposure. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects in every column. SE: HC2 heteroskedasticity-robust standard errors. Sample: included city-level cross-section with one observation per city. Years: outcome changes are measured from 1997 to 2000; controls are measured in 2000. Outcome scale: all outcomes are log changes; no outcome is IHS-transformed or expressed per capita. Placebo check testing whether future predicted exposure is already associated with pre-treatment fiscal growth.

Table A20. Lead Placebo Tests for Fiscal Compensation Timing

	Lead Placebos	
	Grants 97–00	Social Sec. 98–00
SOE employment-share decline, 2000–2007	0.113	0.012
	(0.067)	(0.314)
Region FE	Y	Y
Controls	Y	Y
Cluster	Province	Province
Outcome years	1997–2000	1998–2000
Timing	Lead placebo	Lead placebo
Num. obs.	328	330

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: reported intergovernmental grant receipts and social-security spending use source fiscal-statistics currency units; SOE employment decline is a share change. Transformations: outcomes are log changes, computed as log(y_end + 1) minus log(y_start + 1); SOE employment-share decline is SOE employment share in 2000 minus SOE employment share in 2007, so positive values indicate larger declines. Controls: baseline 2000 share of the population employed in manufacturing, ethnic-minority share, GDP per capita, working-age share, and log out-of-province migrants. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: city-level cross-section with available SOE employment and fiscal outcomes. Years: SOE employment decline is measured from 2000 to 2007; lead-placebo outcomes are changes from 1997 to 2000 for reported grant receipts and 1998 to 2000 for social-security spending. Outcome scale: all outcomes are log changes; no outcome is IHS-transformed or expressed per capita. The 1997 fiscal schedule groups grants more coarsely than the 2000 schedule. Because SOE employment is observed consistently only in 2000 and 2007, the timing columns test whether the later employment decline already predicts pre-2000 grant or social-spending growth rather than estimating a full annual event study.

CTA recruitment records are unavailable before 2002. I therefore present descriptive LTA staffing data for the 13 cities of Hubei Province in Figure A8, where pre-treatment city-level records are available. I find no visually common break around 2002, but the series concern a single province and a different tax administration. Because a comparable city-level regression series is unavailable, I use the figure as descriptive context rather than a formal pre-trend test.

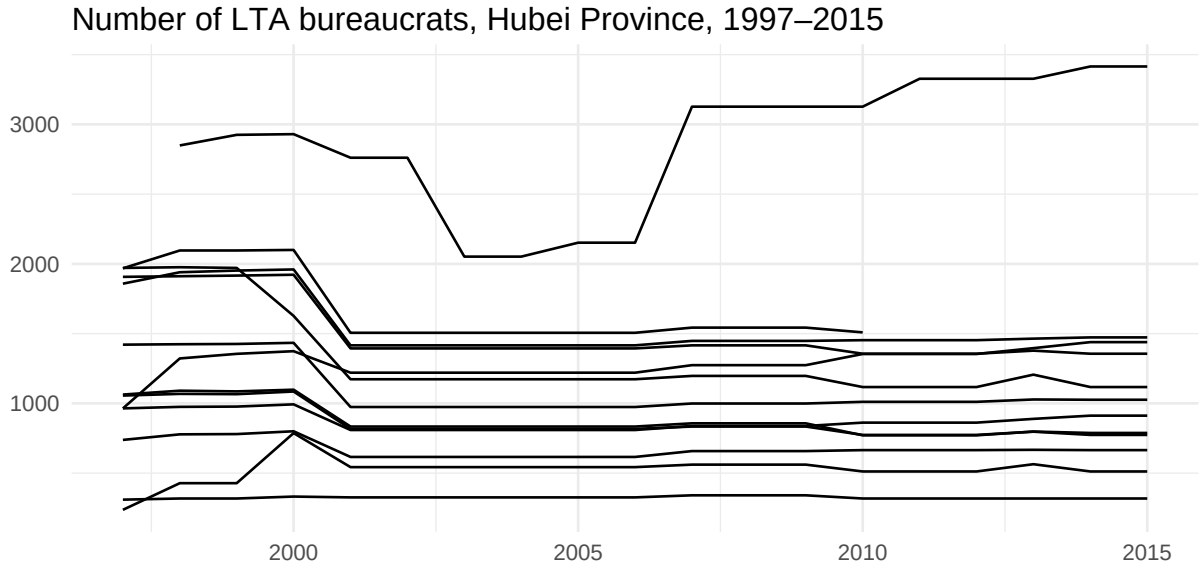


Figure A8. Descriptive LTA staffing series for Hubei cities

Note. This figure shows LTA staffing over time in Hubei Province. Each line represents a city in the province.

H.6 Fiscal Compensation: Measurement and Robustness Checks

For the two-period fiscal compensation checks, I cluster standard errors at the city level. The reported-grant measure harmonizes explicit tax-rebate, general-transfer, special-transfer, and other upper-government subsidy lines across the 2000 and 2007 fiscal schedules. It excludes prior-year balances, inter-fund transfers, other balancing receipts, and the 2007 line combining national-debt on-lending with carryover items. The schedules disaggregate grants differently, so I harmonize the explicitly labeled receipt lines and aggregate city-and-county accounts consistently across years.

Table A21. Reported Grant Receipts and Worker-Linked Outcomes without Controls

	Reported Grant Receipts			Worker-Linked Outcomes	
	Total	Per Capita	Per SOE	Soc. Sec.	Recorded Retirees
SOE employment share (remaining)	-0.413^{***}	-0.501^{***}	-3.564^{***}	-3.300^{***}	-0.101^{*}
	(0.077)	(0.103)	(0.444)	(0.481)	(0.058)
City FE	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y
Controls	N	N	N	N	N
Cluster	City	City	City	City	City
Num. obs.	666	664	666	666	666

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: reported intergovernmental grant receipts and social-security spending use source fiscal-statistics currency units; recorded retirees are counts. Transformations: reported grant receipts are log(y + 1); per-capita grant outcomes subtract log total population; per-SOE outcomes subtract log(SOE employment + 1); social-security and retiree logs are winsorized. Controls: none. FE: city and year fixed effects in every column. SE: standard errors clustered by city. Sample: two-period city panel. Years: panel years are 2000 and 2007. Outcome scale: columns 1 and 5 are logged outcomes; column 2 reports logged grants adjusted for total population; columns 3 and 4 are per-SOE logs; no outcome is IHS-transformed. Reported grant receipts sum fiscal-account lines explicitly labeled as tax rebates, subsidies, or transfers and exclude carryovers, inter-fund transfers-in, and residual financing. The schedules change their component disaggregation across years, and city-and-county totals may include within-prefecture transfers. All columns omit time-varying controls.

The resulting outcome captures the intergovernmental resources reported as received by each local government. Consolidation prevents identifying a unique upper-government tier for every receipt.

Table A21 reports no-control estimates for total, population-normalized, and SOE-worker-normalized grants. It also reports social-security spending per remaining SOE worker and recorded retirees. Per-SOE measures capture resource intensity relative to the remaining workforce and restructuring burdens. Utility spending, other general spending, and public employment remain unrelated to SOE employment-share decline in Table A22. Together, these comparisons locate the response in fiscal compensation and obligations to SOE workers rather than general budget expansion.

Table A22. SOE Employment Share and Other Fiscal Outcomes

	Other Fiscal Outcomes		
	Utility	Other Spend	Public Emp.
SOE employment share (remaining)	−0.190	0.144	0.034
	(0.118)	(0.204)	(0.034)
City FE	Y	Y	Y
Year FE	Y	Y	Y
Controls	Y	Y	Y
Num. obs.	664	662	664

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: spending outcomes use source fiscal-statistics currency units; public employment is a worker count. Transformations: spending and public-employment outcomes are log(y + 1), with the public-employment log winsorized. Controls: share of the population employed in manufacturing, ethnic-minority share, GDP per capita, working-age share, and log out-of-province migrants. FE: city and year fixed effects in every column. SE: standard errors clustered by city. Sample: two-period city panel. Years: panel years are 2000 and 2007. Outcome scale: all outcomes are logged levels; no outcome is IHS-transformed or expressed per capita. The outcomes capture less-direct general public-sector responses. The regressor is SOE employment share (remaining), so negative coefficients indicate larger outcomes in cities with larger SOE employment-share declines.

H.7 Sensitivity, Leverage, and Staffing Composition Checks

Table A23 examines whether extreme cities or administrative centers drive the private-firm and CTA-recruitment estimates. I winsorize the dependent variable and the exposure/instrument pair, exclude provincial capitals and municipalities, and drop observations identified as high leverage in the corresponding covariate-adjusted diagnostic regressions.

Table A23. Outlier Sensitivity and Leverage Checks

	Main 2SLS	Winsor Y	Winsor X/Z	Winsor all	No capital/muni.	Drop leverage
Δ Private firms	8.247***	8.129***	8.122***	8.004***	2.032**	6.126***
	(2.331)	(2.221)	(2.251)	(2.141)	(0.894)	(1.913)
CTA positions	0.075***	0.056***	0.074***	0.056***	0.051***	0.061**
	(0.027)	(0.018)	(0.026)	(0.018)	(0.016)	(0.024)
Estimator	2SLS	2SLS	2SLS	2SLS	2SLS	2SLS
Region FE	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y
Winsor Y	N	Y	N	Y	N	N
Winsor X/Z	N	N	Y	Y	N	N
Sample	All	All	All	All	No capital/muni.	No high leverage

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Units: export exposures are measured in US dollars per thousand yuan of baseline industrial output; private firms and CTA positions are count outcomes. Transformations: all columns estimate the same 2SLS specification as the main headline models; columns 2 and 4 winsorize the dependent variable at the 1st and 99th percentiles; columns 3 and 4 winsorize the raw exposure and tariff-predicted instrument at the 1st and 99th percentiles within the estimation sample. Controls: baseline 2000 census and industry controls: unemployment share, share of the population employed in manufacturing, share of the population aged 20–34, male share, and the first three principal components of baseline industrial structure. FE: regional fixed effects in every column. SE: standard errors clustered by province. Sample: included city-level cross-section; column 5 excludes provincial capitals and municipalities; column 6 excludes high-leverage observations. Years: firm changes are measured from 2000 to 2007; CTA recruitment is measured in 2007; controls are measured in 2000. Outcome scale: private firms and CTA positions are level outcomes; no outcome is logged, IHS-transformed, or expressed per capita. Additional check for outlier sensitivity. High-leverage observations are identified separately for each outcome using hat values from the corresponding OLS design with export exposure, the full control set, and region indicators; the cutoff is 2p/n. Provincial capitals are identified by city-code suffix 01; municipalities are province codes 11, 12, 31, and 50.

Table A24 examines an alternative explanation for recruitment: replacement of retiring personnel. I relate future CTA recruitment to the age and educational composition of provincial tax staff.

Replacement hiring should produce a positive relationship between the share of older tax bureaucrats and subsequent recruitment. A lower share of college-educated staff, more common in older cohorts, should also predict more hiring if the government replaces retiring personnel with younger, more educated recruits. Neither relationship is consistent across the estimates. Replacement needs therefore do not fully account for CTA expansion.

Table A24. Tax Bureaucrat Traits in 2000 and Subsequent CTA Recruitment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Below 25 Share	714.858						
	(1724.461)						
25–35 Share		−181.859					
		(1421.602)					
35–45 Share			−1934.295*				
			(1118.151)				
45–54 Share				615.084			
				(1435.340)			
Above 59 Share					2225.701		
					(2487.137)		
Mean Age						−15.466	
						(44.731)	
College Share							480.015
							(1353.528)
Num. obs.	30	30	30	30	30	30	30

$^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$. Standard errors are robust.